

An efficient quantum algorithm for computing S -units and its applications

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Abstract

In this paper, we provide details on the proofs of the quantum polynomial time algorithm of Biasse and Song (SODA 16) for computing the S -unit group of a number field. This algorithm directly implies polynomial time methods to calculate class groups, S -class groups, relative class group and the unit group, ray class groups, solve the principal ideal problem, solve certain norm equations, and decompose ideal classes in the ideal class group. Additionally, combined with a result of Cramer, Ducas, Peikert and Regev (Eurocrypt 2016), the resolution of the principal ideal problem allows one to find short generators of a principal ideal. Likewise, methods due to Cramer, Ducas and Wesolowski (Eurocrypt 2017) use the resolution of the principal ideal problem and the decomposition of ideal classes to find so-called “mildly short vectors” in ideal lattices of cyclotomic fields.

1 Introduction

Let K be a number field of degree n and $\mathcal{O}$ be an order in K with discriminant Δ . The set of elements $\alpha \in K$ such that $\exists (e_i)_{i \leq |S|} \in \mathbb{Z}^{|S|}$, $(\alpha) = \mathfrak{p}^{e_1} \cdots \mathfrak{p}^{e_{|S|}}$ is a multiplicative group called the S -unit group of K . This notion generalizes the units of $\mathcal{O}$ which are S -units for $S = \emptyset$, and computing the S -unit group is an important task in computational number theory. Most notably it applies to the computation of the ideal class group of $\mathcal{O}$, the resolution of the principal ideal problem in $\mathcal{O}$, and the resolution of norm equations of the form $\mathcal{N}_{L/K}(x) = \theta$ where $\theta \in K$, as shown by Simon [31] and Fieker [17, 19].

The ideal class group $\text{Cl}(\mathcal{O})$ is the finite abelian group consisting of the invertible fractional ideals of $\mathcal{O}$ up to principal factors and has order $|\Delta|^{O(1)}$. Computing the ideal class group is an essential task in number theory that occurs in particular in the resolution of unproven heuristics such as the Cohen-Lenstra heuristics [12] on class groups of quadratic number field, Littlewood’s bounds [27] on $L(1, \chi)$, or Bach’s bound [2] on the maximum norm of the generators required to generate the class group. Besides being a fundamental problem, computing the ideal class group is also strongly connected to number theoretic problems occurring in cryptography. For example, it is at the heart of the only known

unconditional classical subexponential algorithm for integer factorization [26]. Finding relations between elements in $\text{Cl}(\mathcal{O})$ also occurs in curve-based cryptography. Indeed, both classical [6, 24] and quantum [10] subexponential methods for computing isogenies between elliptic curves depend on it.

Given an ideal $\mathfrak{a} \subseteq \mathcal{O}$, deciding whether or not $\mathfrak{a}$ is principal, and if so, finding $\alpha \in \mathcal{O}$ such that $\mathfrak{a} = (\alpha)$ is called the Principal Ideal Problem. It has direct applications to the computation of relative class groups and unit groups, and computing the S -class group of a number field. It is also relevant to lattice-based cryptography, which has received a considerable attention since it allows quantum-safe cryptosystems and homomorphic encryption schemes. For efficiency reasons, there have been many proposals of schemes using lattices arising from ideals in the ring of integers of a number field, and in particular principal ideals generated by a small element (for example, see the homomorphic encryption scheme of Smart and Vercauteren [32] and the multilinear maps of Garg, Gentry and Halevi [20]). It was subsequently proved that solving the principal ideal problem in polynomial time directly induces a polynomial time attack on schemes relying on the hardness of finding the short generator of a principal ideal [13].

Previous work Computing the ideal class group and the unit group is a problem that has been extensively studied in both the classical and quantum setting. Despite these efforts, there are no known polynomial time algorithms for these tasks. On the other hand, there are quantum polynomial time algorithms for several hard computational problems in number theory based on quantum algorithms for the Hidden Subgroup Problem (HSP). Shor showed that integer factorization and the discrete logarithm problem could be solved in polynomial time [30], and Hallgren described a polynomial time algorithm for solving the Pell’s equation [23]. Similar methods were used to compute the class group and the unit group in polynomial time in classes of number fields of fixed degree [22, 29]. The approach of [22] relies on the resolution of the HSP in a bounded and discretized approximation of $\mathbb{R}^m$, which does not seem to apply when the degree of the fields grows to infinity. In a recent breakthrough, Eisenträger, Hallgren, Kitaev and Song [15] described a polynomial time algorithm for computing the unit group in classes of number fields of arbitrary degree. One of the main tools they developed is a continuous HSP definition on $\mathbb{R}^m$ and an efficient quantum algorithm solving it. In essence, their new HSP definition enforces stringent *continuity* properties on the function that hides the subgroup. This makes the function more amenable to quantum Fourier sampling.

Our contribution In this paper, we present a quantum algorithm to compute the S unit group of a number field of arbitrary degree in polynomial time. It readily applies to the computation of the ideal class group and to the resolution of the principal ideal problem, and well as to other related tasks in computational number theory. We follow a different framework than the previous work in constant-degree number fields due to Hallgren [22]. We show that both the ideal class group computation and PIP reduce to a more general problem of computing the S -unit group for suitable set of prime ideals S . For example, for the ideal class group computation S is chosen to be a succinct generating set of $\text{Cl}(\mathcal{O})$. Then we give an efficient quantum algorithm for computing the S -unit group by extending the work by Eisenträger, Hallgren, Kitaev and Song [15]. We show an efficient quantum

reduction from the S -unit group problem to HSP on $\mathbb{R}^m$ as defined in [15], which then can be solved efficiently by the quantum HSP algorithm in [15]. We also show how to get exact compact representations of the desired field elements with respect to a given integral basis for $\mathcal{O}$, while [15] only returns fixed point rational approximations of the units. Compact representations are usually easier for further algebraic processing. Our main results are summarized in the next theorem.

Theorem 1 (*S -unit group computation*). *There is a quantum algorithm for computing the S -unit group of a number field K in compact representation which runs in polynomial time in the parameters $n = \deg(K)$, $\log(|\Delta|)$, $|S|$ and $\max_{\mathfrak{p} \in S} \{\log(\mathcal{N}(\mathfrak{p}))\}$, where Δ is the discriminant of the ring of integers of K .*

Corollary 1. *There are quantum polynomial time algorithms for the resolution of the following tasks in computational number theory:*

- *Ideal class group computation (under GRH),*
- *S -class group computation (under GRH),*
- *Relative class group and unit group computation (under GRH),*
- *Ideal class decomposition in the ideal class group (under GRH),*
- *Principal Ideal Problem,*
- *Ray class group computation (under GRH),*
- *Norm equation resolution,*

where GRH denotes the Generalized Riemann Hypothesis.

As an important corollary, combining recent works in lattice cryptanalysis [9, 13], our results induce a quantum polynomial-time attack on an entire family of cryptosystems relying on the hardness of finding a short generator of a principal ideal.

Response to a recent preprint from de Boer and Felderhoff [14] In a preprint published on Oct. 3rd 2025 and updated on Oct. 22nd 2025, K. de Boer and J. Felderhoff presented a similar quantum algorithm as our 2016 work [5], which combined with [15] allows one to compute the S -unit group of a number field in quantum polynomial time. The work of de Boer and Felderhoff [14] uses the same strategy as our previous work [5] with some technical differences. They claimed that the detailed analysis of the degrees of the polynomial dependencies in the work of Biasse-Song [5] (which itself relies on Eisentraeger et al. [15, 16]) was “not currently possible”. They additionally suggested that the S -unit algorithm of [5] did not run in polynomial time [14, p. 5]. Finally, de Boer and Felderhoff claimed that the key differences between their algorithm offered an advantage compared to the strategy of [5]. Below, we comment on these statements.

First and foremost, we would like to confirm that the 2016 paper of Biasse and Song to compute S -unit groups [5] does indeed run in polynomial time. There is an unfortunate typo in the published version of [5, Th. 5.1] which was copied to an earlier version of Theorem 2

of this document. It showed a dependency that appeared to be polynomial in the bound of some input coefficients instead of being polynomial in their bit size. The former obviously does not qualify as a polynomial dependency in the size of the input. We apologize for this confusion. Note that the rest of [5], as well as the previous version of this document remained consistent with a dependency in the bit size (the paragraph above Theorem 2 actually made a claim of polynomial behavior in the bit size of the input even in older versions of this document). It is achieved through an ideal exponentiation strategy devised by Eisentraeger et al. [15], and used by Biasse-Song [5]. In this document, it is explained in details in Section 4.1. While we are sorry that such a confusing typo got included in the published version of our 2016 work [5], we would like to point out that we had multiple email exchanges with de Boer and Felderhoff during the month preceding the release of their preprint. They never mentioned their concerns regarding the run time of our algorithm, and chose to publish claims of the exponential complexity without seeking our input.

Second, we would like to state that prior work that established the quantum polynomial time of the unit and S -unit quantum algorithm were published at STOC 2014 [15] and SODA 2016 [5] respectively. Due to the strict page limits of these venues, the choice was made to not specify the exact degrees of the polynomial dependencies in the input parameters of these algorithms. However, this does not mean that it “not currently possible”, or that it is even technically challenging. It is a tedious but straightforward process that can be achieved with known methods. We added such analysis in this updated draft. It mostly relies on techniques to control the precision of LLL reductions on fixed points approximations of vectors originally described in 1987 by buchmann and Pohst [8], and later adapted by Eisentrager et al. [15].

Finally, we comment on the advantage offered by the modified S -unit oracle of de Boer and Federhoff. Their main proposal is to precompute some of these powers in ideal multiplication and exponentiation. However, the savings obtained are moderate, and actually turn into a loss if we focus on cases of interest to the cryptography community (for example when $|S| = 1$, which is a case that allows the resolution of the Principal Ideal Problem). Indeed, while fewer LLL reductions are required in the oracle described in [14], the size of the entries involved is significantly larger. Indeed, our quantum oracle always performs multiplications between ideals of norm 1 (following a method already presented in Eisentraeger et al. [15]). On the other hand, in [14], the norm of the ideals is non trivial and grows over the course of the algorithm. In the end, the dependency in the degree n in the bit size β of the vectors given to the LLL solver is in n^7 for [14] whereas it is only n^5 for our method. Then, the gate count of the LLL resolution has a dependency in $\beta^{3.5}$, which induces a penalty of n^7 . Since on the other hand our method requires more LLL resolutions over larger matrices, the final dependency in n of the gate count of the oracle described by de Boer and Federhoff is in $n^{31.5}$, while ours is in $n^{35.5}$. However, this slight advantage vanishes once we account for the additional restriction in the work of de Boer and Federhoff that consists in assuming that the set of primes S generates the ideal class group of the field. If S is too small, they suggest enlarging it so that it contains all primes of norm up to $12 \log^2 |\Delta|$ where Δ is the discriminant of the field (under the Generalized Rieman Hypothesis, such a set of primes generate the ideal class group [2]). The apparent issue of this workaround is that in the important case of the resolution of the Principal Ideal Problem (which can be reduced in polynomial time to the search of S -units where $|S| = 1$),

this induces a penalty of $|S|^{17.5} \in O((\log|\Delta|)^{35})$ in the complexity of the gate cost. In cyclotomic fields, where $\log|\Delta| \in \Theta(n \log(x))$, this means that the degree of the highest term in the degree becomes n^{49} , which is significantly larger than the $n^{35.5}$ dependency of the gate count of our quantum oracle.

2 Technical background

In this section we review some useful background in number theory and introduce some definitions and notations. The notions of ideal class group and S -unit group are standard, and can be found in many books. We suggest Neukirch's book [28] for the fundamental aspects of this theory and Cohen's book [11] for the algorithmic aspects. We invite the reader who is already familiar to these topics to pay attention to the non-standard notion of E -ideal that we introduce in the following.

2.1 Number Theory

Number fields A number field K is a finite extension of $\mathbb{Q}$. Its ring of integers $\mathcal{O}_K$ has the structure of a $\mathbb{Z}$ -lattice of degree $n = [K : \mathbb{Q}]$, and the orders $\mathcal{O} \subseteq \mathcal{O}_K$ are the sublattices of $\mathcal{O}_K$ which have degree n and which are equipped with a ring structure. Throughout this paper, we assume that $\mathcal{O}$ is an order in a number field K , and we denote by $\omega_1, \dots, \omega_n$ a $\mathbb{Z}$ -basis, that is $\mathcal{O} = \mathbb{Z}\omega_1 \oplus \dots \oplus \mathbb{Z}\omega_n$. A number field has n_1 real embeddings and n_2 pairs of complex embeddings which we denote $(\sigma_j : K \rightarrow \mathbb{R})_{j \leq n_1}, ((\sigma_j, \bar{\sigma}_j) : K \rightarrow \mathbb{C})_{j \leq n_2}$ with $n_1 + n_2 = n = \deg(K)$. These embeddings define two essential maps, namely the norm and trace maps which are given by $\mathcal{T}(x) := \sum_{\sigma} \sigma(x) \in \mathbb{Q}$ and $\mathcal{N}(x) := \prod_{\sigma} \sigma(x) \in \mathbb{Q}$. The trace map is additive while the norm map is multiplicative. Note that $\mathcal{T}(\mathcal{O}) \subseteq \mathbb{Z}$ and $\mathcal{N}(\mathcal{O}) \subseteq \mathbb{Z}$. We measure the size of the ring $\mathcal{O}$ by $\log|\Delta|$ where $\Delta := (\det(\sigma_j(\omega_k)))^2$ is its discriminant, and it equals the volume of the fundamental domain of $\mathcal{O}$. Equivalently, the discriminant can be defined from the trace map by $\Delta := \det(\mathcal{T}(\omega_i \omega_j))_{i,j \leq n}$.

The ideal class group The fractional ideals of $\mathcal{O}$ generalize the notion of ring ideals of $\mathcal{O}$. They are the subsets of K of the form $\mathfrak{a} = \frac{1}{d}I$ where $d \in \mathbb{Z}^+$ and $I \subseteq \mathcal{O}$ is an (integral) ideal of $\mathcal{O}$. A fractional ideal $\mathfrak{a}$ is invertible if $\mathfrak{a}^{-1} := \{x \in K \mid x\mathfrak{a} \subseteq \mathcal{O}\}$ is also a fractional ideal. The invertible fractional ideals have a multiplicative group structure, and the principal fractional ideals are one of its subgroups. The ideal class group is defined by

$$\text{Cl}(\mathcal{O}) := \mathcal{I}/\mathcal{P},$$

where $\mathcal{I}$ is the multiplicative group of fractional invertible ideals of $\mathcal{O}$ and $\mathcal{P}$ is the subgroup of elements of $\mathcal{I}$ that are principal. This means that we identify $\mathfrak{a}$ and $\mathfrak{b}$ in $\text{Cl}(\mathcal{O})$ if there is $\alpha \in K$ such that $\mathfrak{a} = (\alpha)\mathfrak{b}$. Ideals are sublattices of $\mathcal{O}$ of rank n , and we define their norm by $\mathcal{N}(I) := |\mathcal{O}/I|$. This notion naturally extends to fractional ideals using the multiplicative rule $\mathcal{N}(\mathfrak{a}/\mathfrak{b}) := \mathcal{N}(\mathfrak{a})/\mathcal{N}(\mathfrak{b})$. This notion of norm extends the norm on K in the sense that if $\mathfrak{a} = (\alpha)$, then $\mathcal{N}(\mathfrak{a}) = \mathcal{N}(\alpha)$.

The S -unit group The S -units are a generalization of the units $\mathcal{O}^*$, which are the invertible elements of $\mathcal{O}$. The unit group can alternatively be defined as the $\alpha \in \mathcal{O}$ with $|\mathcal{N}(\alpha)| = 1$, or the $\alpha \in \mathcal{O}$ such that $(\alpha) = \mathcal{O}$. The unit group $\mathcal{O}^*$ satisfies $\mathcal{O}^* \simeq \mu \times \langle \varepsilon_1 \rangle \times \dots \times \langle \varepsilon_r \rangle$, where $r := n_1 + n_2 - 1$, μ is the set of roots of unity and the ε_i are torsion-free units. Let $S = \{\mathfrak{p}_i\}$ be a finite set of prime ideals of $\mathcal{O}$, the S -units are the elements $\alpha \in K$ such that there is $(v_i(\alpha))_{i \leq |S|} \in \mathbb{Z}^{|S|}$ with $(\alpha) = \mathfrak{p}_1^{v_1(\alpha)} \dots \mathfrak{p}_{|S|}^{v_{|S|}(\alpha)}$. Note that the S -units are elements of K . They form a multiplicative group $U(S)$ satisfying $U(S) \simeq \mu \times \langle \varepsilon_1 \rangle \times \dots \times \langle \varepsilon_{r+|S|} \rangle$, where $r := n_1 + n_2 - 1$, μ is the set of roots of unity and the ε_i are torsion-free S -units.

E -ideals The number field K can be naturally embedded into $E := \mathbb{R}^{n_1} \times \mathbb{C}^{n_2}$ by setting $z \in \mathcal{O} \mapsto (\sigma_1(z), \dots, \sigma_{n_1+n_2}(z))$. As in [15], we denote by $\underline{\mathcal{O}}$ the image of $\mathcal{O}$ via this embedding. The set $\underline{\mathcal{O}}$ inherits from the lattice structure of $\mathcal{O}$, i.e. it can be identified as a lattice in $\mathbb{R}^n$, as well as from the multiplication between elements (which is performed component-wise). The image of the fractional ideals of K in E are lattices $\Lambda \subseteq E$ with the property that $x\Lambda \subseteq \Lambda$ for all $x \in \underline{\mathcal{O}}$. We define the E -ideals as all the lattices in E satisfying this property. When there is no ambiguity, we identify a fractional ideal of $\mathcal{O}$ and the corresponding E -ideal.

Definition 1 (E -ideals). *Let $E := \mathbb{R}^{n_1} \times \mathbb{C}^{n_2}$ and $\underline{\mathcal{O}}$ the image of $\mathcal{O}$ via the embedding $K \rightarrow E$. An E -ideal is a lattice $\Lambda \subseteq E$ such that $\forall x \in \underline{\mathcal{O}}, x\Lambda \subseteq \Lambda$.*

2.2 HSP resolution

Continuous HSP We review the definition of continuous HSP proposed by Eisenträger et al. [15], for which they have shown an efficient quantum algorithm.

Definition 2 (Continuous HSP over $\mathbb{R}^m$). *The unknown subgroup $L \subseteq \mathbb{R}^m$ is a full-rank lattice satisfying some promise: the norm of the shortest vector is at least λ and the unit cell volume is at most d . The oracle has parameters (a, r, ε) . Let $f : \mathbb{R}^m \rightarrow \mathcal{S}$ be a function, where $\mathcal{S}$ is the set of unit vectors in some Hilbert space. We assume that f hides L in the following way.*

1. f is periodic on L , i.e. $f(x) = f(x + v)$ for all $x \in \mathbb{R}^m$ and $v \in L$;
2. f is Lipschitz with constant a , i.e. $\|f(x) - f(y)\| \leq a\|x - y\|$ for all $x, y \in \mathbb{R}^m$;
3. If the distance between the cosets $(x \bmod L)$ and $(y \bmod L)$ is greater or equal to r , i.e. if $\min_{v \in L} \|x - y - v\| \geq r$, then $|\langle f(x) | f(y) \rangle| \leq \varepsilon$.

Under these conditions, the problem is to compute a basis of L by a quantum algorithm that can make oracle calls $|x\rangle \mapsto |x\rangle \otimes |f(x)\rangle$.

Actually, the definition also applies more generally to other topological groups $G = \mathbb{R}^k / \Lambda \times D$ with a proper metric on G [15, Sect.6.1]. Here G is decomposed to a continuous part, which is the quotient of $\mathbb{R}^k$ over some lattice Λ , and a discrete part that is finitely generated. It is nonetheless sufficient to consider HSP on $\mathbb{R}^m$, because the more general case can be reduced to HSP on $\mathbb{R}^m$ [15], and hence can be solved efficiently. In the following,

we define a control group G on which a first version of our HSP oracle will be defined. We prove HSP properties on G , and then extend it to $\mathbb{R}^m$.

Suppose $\sigma_1, \dots, \sigma_{n_1}$ are the real embeddings of K , and that $\sigma_{n_1+1}, \dots, \sigma_{n_1+n_2}$ are the (non-pairwise conjugate) complex embeddings of K . Assume also that $S = \{\mathfrak{p}_1, \dots, \mathfrak{p}_s\}$ where $\mathcal{N}(\mathfrak{p}_i) = p_i^{e_i}$. An element $x \in U_S$ satisfies $\prod_{i=1}^{n_1+2n_2} \sigma_i(x) = \mathcal{N}(x) = \prod_i p_i^{e_i v_i(x)}$. This means that we know that

$$\log |\sigma_1(x)| = - \sum_{i=2}^{n_1} \log |\sigma_i(x)| - 2 \sum_{i=n_1+1}^{n_2} \log |\sigma_i(x)| + \sum_{i \leq s} e_i v_i(x) \log p_i.$$

Therefore, $x \in U_S$ corresponding to $(x_1, \dots, x_n) = (\sigma_1(x), \dots, \sigma_n(x)) \in \mathbb{R}^{n_1} \times \mathbb{C}^{n_2}$ is uniquely identified by the element $x^G \in G := \mathbb{R}^{n_1+n_2-1} \times \mathbb{Z}_2^{n_1} \times (\mathbb{R}/\mathbb{Z})^{n_2} \times \mathbb{Z}^s$ where

- $x_i^G = \log(|x_{i+1}|)$ for $1 \leq i < n_1 + n_2$,
- $x_i^G = \delta_i \in \mathbb{Z}_2$ where $x_{i-n_1-n_2+1} = (-1)^{\delta_i} |x_{i-n_1-n_2+1}|$ for $n_1 + n_2 \leq i < 2n_1 + n_2$.
- $x_i^G = \theta_i \in \mathbb{R}/\mathbb{Z}$ where $x_{i-2n_1-n_2+1} = e^{2i\pi\theta_i} |x_{i-2n_1-n_2+1}|$ for $2n_1 + n_2 \leq i < 2n_1 + 2n_2$.
- $x_i^G = v_{\mathfrak{p}_{i-2n_1+2n_2}}(x)$ for $2n_1 + 2n_2 \leq i < 2n_1 + 2n_2 + s$.

Conversely, we have a map $\phi : G \rightarrow \mathbb{R}^{n_1} \times \mathbb{C}^{n_2}$ such that $\phi(x^G) = x$ by choosing $|x_i| = e^{x_{i+1}^G}$ for $1 \leq i < n_1 + n_2$, and

$$|x_1| = \frac{\prod_{i=1}^s p_i^{e_i x_{i+2n_1+2n_2}^G}}{\prod_{i=2}^{n_1} |x_i| \prod_{i=n_1+1}^{n_1+n_2} |x_i|^2}. \quad (1)$$

Then we do

- $x_i \leftarrow (-1)^{x_{i+x_1+x_2-1}^G} |x_i|$ for $1 \leq i \leq n_1$ and
- $x_i \leftarrow e^{2i\pi x_{i+2n_1+2n_2-1}^G} |x_i|$ for $n_1 < i \leq n_1 + n_2$.

Definition 3 (Control group G). *Let K be a number field of signature (n_1, n_2) , and S a set of primes above $(p_i)_{i \leq s}$. We define the following groups:*

- $G = \mathbb{R}^{n_1+n_2-1} \times \mathbb{Z}_2^{n_1} \times (\mathbb{R}/\mathbb{Z})^{n_2} \times \mathbb{Z}^s$ the control group.
- $L = U_S^G \subseteq G$ the image of the S -unit group of K , which is a lattice.

The map ϕ is readily extended beyond elements of G that correspond to an S -unit. In this case, $\phi(u, v) \in \mathbb{R}^{n_1} \times \mathbb{C}^{n_2}$ for $u \in \mathbb{R}^{n_1+n_2-1} \times \mathbb{Z}_2^{n_1} \times (\mathbb{R}/\mathbb{Z})^{n_2}$ and $v \in \mathbb{Z}^s$ does not necessarily correspond to an element $x \in K$ with $\mathcal{N}(x) = \prod_i p_i^{e_i v_i}$. On the other hand, in general, there is no canonical way to map an element of $\mathbb{R}^{n_1} \times \mathbb{C}^{n_2}$ that is not an S -unit to an element of G .

The control group can be seen as the projection of $\tilde{G} = \mathbb{R}^k \times \mathbb{Z}^l$ where

- $k = n_1 + 2n_2 - 1$.
- $l = n_1 + s$.

We denote by $\gamma : \tilde{G} \rightarrow G$ the projection map, and by $\tilde{L} \subseteq \tilde{G}$ the pre-image of L by γ . It is a lattice in $\tilde{G}$. We also construct an oracle $g = f_q \circ f_c : G/L \rightarrow \mathcal{H}$ where

- $f_c(t, v) = e^{\mathbf{t}} \underline{\mathcal{Q}} \prod_{\mathbf{p} \in S} \mathbf{p}^{-v_i}$, which is a lattice in $E := \mathbb{R}^{n_1} \times \mathbb{C}^{n_2}$.
- $f_q(L_E) = |L_E\rangle := \gamma \sum_{v \in L_E} g_s(v) |\text{str}_{\nu, n}(v)\rangle$ which is a quantum state (see Section 4.4 for a definition of the straddle encoding $|\text{str}_{\nu, n}(v)\rangle$ introduced in [15]).

To prove the HSP properties of $f : G \rightarrow \mathcal{H}$, we need a notion of distance between ideals of $E = \mathbb{R}^{n_1} \times \mathbb{C}^{n_2}$. An ideal in E is a lattice that is stable by multiplication by elements in $\underline{\mathcal{Q}}$ (the embedding of $\mathcal{O}$ in E). We deal with elements in E by embedding them in $\mathbb{R}^n = \mathbb{R}^{n_1+2n_2}$ (via $z \in \mathbb{C} \mapsto \Re(z), \Im(z)$). Each E -ideal $\mathcal{L}$ can be defined by a matrix $M_{\mathcal{L}} \in \mathbb{R}^{n \times n}$ whose rows are a $\mathbb{Z}$ -basis of $\mathcal{L}$. Note that E -ideals $\mathcal{L}, \mathcal{L}'$ can be multiplied, but $M_{\mathcal{L}\mathcal{L}'}$ is in general *not* equal to $M_{\mathcal{L}}M_{\mathcal{L}'}$.

Notation. *The Euclidean norm is used in different spaces. When there is a potential ambiguity, we use a subscript to specify the space. More specifically, suppose there is a group H and s, t such that $\alpha : H \hookrightarrow \mathbb{R}^s \times \mathbb{C}^t$, then for $x \in H$, we denote by $\|x\|_H = \|\alpha(x)\|$, i.e. if $x = (x_1, \dots, x_{s+t})$, then $\|x\|_H = \sqrt{\sum_{i \leq s} |x_i|^2 + 2 \sum_{i > s} |x_i|^2}$.*

Definition 4 (Matrix distance between E -ideals). *Let $\mathcal{L}, \mathcal{L}'$ be two E -ideals. We define the matrix distance between $\mathcal{L}$ and $\mathcal{L}'$ by*

$$\text{dist}(\mathcal{L}, \mathcal{L}') = \inf_{A, M_{\mathcal{L}}, M_{\mathcal{L}'}} \{ \|A\|_2 : M_{\mathcal{L}} = M_{\mathcal{L}'} e^A, A \in \text{Gl}_n(\mathbb{R}) \}$$

As in [16], given an element $x \in E$, we define the matrix $\text{diag}(x) \in \mathbb{R}^{n \times n}$ which is *not* exactly a diagonal matrix.

$$\text{diag}(x) := \begin{pmatrix} x_1 & & & & \\ & \ddots & & & \\ & & x_{n_1} & & \\ & & & \Xi(x_{n_1+1}) & \\ & & & & \ddots \\ & & & & & \Xi(x_{n_2}) \end{pmatrix} \quad \text{where } \Xi(z) := \begin{pmatrix} \Re(z) & -\Im(z) \\ \Im(z) & \Re(z) \end{pmatrix}.$$

Given $x \in E$, the above matrix has the important property that $M_{(x) \cdot \mathcal{L}} = M_{\mathcal{L}} \cdot \text{diag}(x)$ where $\mathcal{L}$ is an E -ideal, and (x) denotes the E -ideal $x \cdot \underline{\mathcal{Q}}$ (a principal ideal generated by x). This is a case where ideal multiplication corresponds to a product of matrices (although $\text{diag}(x)$ is not $M_{(x)}$).

Lemma 1. *Matrices of the form $\text{diag}(x)$ have the following properties:*

1. $\forall x_1, x_2 \in E, \text{diag}(x_1) + \text{diag}(x_2) = \text{diag}(x_1 + x_2)$.
2. $\forall x_1, x_2 \in E, \text{diag}(x_1) \cdot \text{diag}(x_2) = \text{diag}(x_1 \cdot x_2)$.
3. $\forall x \in E, e^{\text{diag}(x)} = \text{diag}(e^x)$ where $e^x = (e^{x_1}, \dots, e^{x_{n_1+n_2}})$.

4. $\forall x \in E$, if $\|\text{diag}(x) - I\| < 1$, then $\log(\text{diag}(x)) = \text{diag}(\log(x))$ where $\log(x) = (\log(x_1), \dots, \log(x_{n_1+n_2}))$.

Proof. For 1) and 2), it suffices to check that $\forall z, z' \in \mathbb{C}$, $\Xi(z) + \Xi(z') = \Xi(z + z')$, and $\Xi(z) \cdot \Xi(z') = \Xi(zz')$. Then, since $e^A = \sum_{k=0}^{\infty} \frac{A^k}{k!}$, we have

$$e^{\text{diag}(x)} = \sum_{k=0}^{\infty} \frac{\text{diag}(x)^k}{k!} = \text{diag} \left(\sum_{k=0}^{\infty} \frac{x^k}{k!} \right) = \text{diag}(e^x).$$

Likewise, to prove 4), we simply use the convergence of $\sum_{k=1}^{\infty} (-1)^{k+1} \frac{(B-I)^k}{k}$ to $\log(B)$ whenever $\|B - I\| < 1$. $\square$

3 High level overview

Our algorithms for the Class Group Problem (CGP) and the Principal Ideal Problem (PIP) consist of reductions to the continuous hidden subgroup problem in two steps, and invoking the quantum HSP algorithm [15] at the end.

$$\begin{aligned} \text{CGP} &\leq_C S_{\text{CGP-units}} \leq_Q \text{HSP}(\mathbb{R}^{O(n)}), \\ \text{PIP} &\leq_Q S_{\text{PIP-units}} \leq_Q \text{HSP}(\mathbb{R}^{O(n)}). \end{aligned}$$

Specifically, we first reduce them to S -unit problems with proper choices of S , which are almost entirely *classical* except that we apply a quantum algorithm for factoring ideals in the case of PIP¹. We describe these reductions to S -units problems in Sect. ???. Next we show a *quantum* reduction from S -units problem for any S to $\text{HSP}(\mathbb{R}^m)$, with $m = O(|S|, n)$. This is the main technical contribution of this work and it generalizes the reduction from (ordinary) unit-group problem to HSP by Eisenträger et al. [15]. The details will appear in Section 8, and we give an overview below.

Given $S = \{\mathfrak{p}_1, \dots, \mathfrak{p}_k\}$, we want to establish a function that hides the S -unit group according to Definition 2. To warm up, we review the reduction for the ordinary unit group (i.e., $S = \emptyset$) [15].

Review: reduction for unit-group [15] Observe that the unit group can be identified as a subgroup of $G := \mathbb{R}^{n_1+n_2} \times \mathbb{Z}_2^{n_1} \times (\mathbb{R}/\mathbb{Z})^{n_2}$, and the mapping

$$\begin{aligned} \varphi : (u_1, \dots, u_{n_1+n_2}, \mu_1, \dots, \mu_{n_1}, \theta_1, \dots, \theta_{n_2}) \\ \mapsto (\dots, (-1)^{\mu_i} e^{u_i}, \dots, e^{2\pi i \theta_i} e^{u_i}, \dots). \end{aligned}$$

translates between the so-called *log coordinates* and the conjugate vector representation. To see this, note that under canonical embeddings, any $z \in \mathcal{O}$ has the conjugate vector representation $(\dots, \sigma_i(z), \dots) \in \mathbb{R}^{n_1} \times \mathbb{C}^{n_2}$. If in addition z is invertible, then $\sigma_i(z) \neq 0$.

¹These reductions are straightforward. But classical algorithms typically compute the S -unit group by solving CGP and solving instances of PIP first. Our quantum algorithm tackles these problems in the reverse order.

Therefore, we can write $\sigma_i(z) = (-1)^{\mu_i} e^{u_i}$ with $\mu_i \in \mathbb{Z}_2$ and $u_i \in \mathbb{R}$ if σ_i is real, or $\sigma_i(z) = e^{2\pi i \theta_i} e^{u_i}$ with $\theta_i \in \mathbb{R}/\mathbb{Z}$ and $u_i \in \mathbb{R}$ if σ_i is complex.

Now one defines f in [15] as composition of two mappings:

$$f : G \xrightarrow{g} \{E\text{-ideals}\} \xrightarrow{f_q} \{\text{quantum states}\}.$$

Given $x \in G$, $g(x) := \varphi(x)\underline{\mathcal{O}} \subseteq E$ produces an E -ideal which is a transformed lattice of $\underline{\mathcal{O}}$. This is motivated by the fact that $\alpha\mathcal{O} = \mathcal{O}$ for any unit $\alpha \in \mathcal{O}^*$. Actually, one can verify easily that $g(x) = g(y)$ iff. $\varphi(x - y) \in \mathcal{O}^*$. Namely g is periodic on $\mathcal{O}^*$. For lacking of a canonical basis to represent real-valued lattices uniquely, which is needed to apply the quantum HSP algorithm, a quantum mapping f_q follows. It encodes a lattice L into a quantum state $|L\rangle$ that is roughly composed of quantum superposition over all lattice points, and hence provides a canonical representation for lattices. We will give more details of the quantum encoding in Sect. 4.

Very informally, one can show that small shift on an input to g causes small variance on the output lattice, but two inputs that are far apart modulo any unit will be mapped to lattices that have small overlap. Moreover, f_q preserves the “closeness” of lattices. Namely, quantum encodings of two lattices will have substantial inner product if and only if the lattices are very well lined up. To formalize these statements and thus proving the HSP properties, nonetheless, turn out to be highly non-trivial. It involves for example defining proper distance measures on various input and output spaces, and analyzing the continuity properties of f with respect to these metrics. This has been a great amount of efforts in [15] with further details in [16]

Other than these analytic properties, to make an efficient reduction, one needs to implement $f = f_q \circ g$ efficiently. In fact, f_q can be implemented efficiently on a quantum computer by standard techniques. Computing g , on the other hand, is much more tricky. For instance e^{u_i} will involve doubly-exponential numbers if we manipulate them naively. Instead one splits the computation into small pieces, in the spirit of repeated squaring, and carefully controls the precision. There is one key observation that guarantees that the size of any intermediate step does not blow up. That is $\mathcal{N}(z) = \pm 1$ for any unit z and hence $\prod_{i=1}^{n_1} e^{u_i} \prod_{j=1}^{n_2} e^{2u_{n_1+j}} = 1$. This indicates one redundant coordinate, and we can hence restrict f on $\mathbb{R}^{n_1+n_2-1} \times \mathbb{Z}_2^{n_1} \times (\mathbb{R}/\mathbb{Z})^{n_2}$ instead. This characterization is also essential to show a suitable bound on the volume of the unit cell of $\mathcal{O}^*$.

Reducing S -units to HSP It is now easier to describe our generalized reduction for S -units. Let $S = \{\mathfrak{p}_1, \dots, \mathfrak{p}_k\}$. By definition, if $\alpha \in \mathcal{O}$ is an S -unit, we have

$$\alpha \cdot \mathcal{O} \cdot \mathfrak{p}_1^{-v_{\mathfrak{p}_1}(\alpha)} \dots \mathfrak{p}_k^{-v_{\mathfrak{p}_k}(\alpha)} = \mathcal{O},$$

where $v_{\mathfrak{p}}(\alpha)$ is the coefficient of $\mathfrak{p}$ in the power of $(\alpha)\mathcal{O}$ (the valuation of α at $\mathfrak{p}$). Therefore the group of S -units U_S corresponds to the subgroup of $G = \mathbb{R}^{n_1+n_2-1} \times \mathbb{Z}_2^{n_1} \times (\mathbb{R}/\mathbb{Z})^{n_2} \times \mathbb{Z}^S$ such that $\phi(y, v) \cdot \underline{\mathcal{O}} \cdot \mathfrak{p}_1^{-v_1} \dots \mathfrak{p}_{|S|}^{-v_{|S|}} = \underline{\mathcal{O}}$. This motivates us to define the function $f_c : G \rightarrow \{E\text{-ideals}\}$ by:

$$f_c : (y, v_1, \dots, v_{|S|}) \mapsto \phi(y, v) \cdot \underline{\mathcal{O}} \cdot \mathfrak{p}_1^{-v_1} \dots \mathfrak{p}_{|S|}^{-v_{|S|}}.$$

We can show that $\hat{g}$ is periodic on U_S . We then apply the same quantum encoding f_q on the output of $\hat{g}$. Namely, our oracle function behaves like:

$$f : G \xrightarrow{f_c} \{E\text{-ideals}\} \xrightarrow{f_q} \{\text{quantum states}\}.$$

While the classical mappings g and f_c bear some similar motivation and we reuse f_q , to prove HSP properties of our function f is not straightforward. We need to define new metrics tailored to the specific group structure that the S -units belong and the E -ideals (lattices in $\mathbb{R}^n$) that our f_c may possibly generate. Then we show quantitatively that under these metrics, small variance in inputs induces slightly perturbed lattices, whereas large variance of inputs modulo any S -units will induce with high fraction of mismatch. Finally we relate the new metrics to the analysis of [15] and conclude the HSP properties. We further extend the function f to obtain an HSP instance on $\mathbb{R}^m$ and work out the necessary bounds (λ, d) as required, which allows us to invoke the quantum HSP algorithm to recover U_S .

4 Defining the oracle function $(\mathbf{y}, \mathbf{v}) \mapsto |\varphi(\mathbf{y})\mathcal{Q}\prod_{\mathfrak{p}\in S}\mathfrak{p}^{-v_i}\rangle$

Our algorithm relies on a classical oracle that takes an element in G and maps it to

$$f_c(y, v_1, \dots, v_{|S|}) = \phi(y, v) \cdot \mathcal{Q}\mathfrak{p}_1^{-v_1} \dots \mathfrak{p}_{|S|}^{-v_{|S|}}.$$

Then the corresponding lattice is encoded by an approximation of the superposition of all its points denoted by f_q . As $G = \mathbb{R}^{n_1+n_2-1} \times \mathbb{Z}_2^{n_1} \times (\mathbb{R}/\mathbb{Z})^{n_2} \times \mathbb{Z}^{|S|}$, we need to work with approximations of real numbers. To perform the necessary arithmetic operations between E -ideals presented in Section 4.2, we use the results of Buchmann and Pohst [8] and of Buchmann and Kessler [7] which rely on fixed point approximations. More specifically, they use the rounding of the 2-adic expansion of real numbers. The approximation of $a \in \mathbb{R}$ of precision $q \in \mathbb{Z}_{>0}$ is $\hat{a} \in \mathbb{Z}$ such that $\left| \frac{\hat{a}}{2^q} - a \right| \leq \frac{1}{2^{q+1}}$. However, it seems that this notion of approximation is not stable when we multiply two approximate numbers together. We made a slight adjustment to their claims to incorporate the case of approximations such that $\left| \frac{\hat{a}}{2^{q_0}} - a \right| \leq \frac{1}{2^q}$ for some $q_0 \geq q$. Then in Section 4.3 we show that the classical oracle runs in polynomial time with respect to the size of the input.

4.1 Splitting up the computation

Let $(y, v_1, \dots, v_{|S|}) \in \mathbb{R}^{n_1+n_2-1} \times \mathbb{Z}_2^{n_1} \times (\mathbb{R}/\mathbb{Z})^{n_2} \times \mathbb{Z}^{|S|}$. The naive computation of

$$f_c(y, v_1, \dots, v_{|S|}) = \phi(y, v) \cdot \mathcal{Q} \cdot \mathfrak{p}_1^{-v_1} \dots \mathfrak{p}_{|S|}^{-v_{|S|}}$$

involves computing $(e^{u_i})_{i \leq n_1+n_2}$, where $y = (u_1, \dots, n_{n_1+n_2}, \theta)$ and u_1 is computed by the rule given by (1) with a phase $\theta \in \mathbb{Z}_2^{n_1} \times (\mathbb{R}/\mathbb{Z})^{n_2}$. Any rational approximation of e^{u_i} has at least $\lceil \log_2(e^{u_i}) \rceil \in O(u_i)$ bits where $\log_2$ denotes the base 2 logarithm. As this is exponential in the bit size of the entry, we need to proceed differently to evaluate f_c . The authors of [15] described a way to split up the computation ensuring that we only manipulate values of

polynomial size. We adapt this method to our specific classical oracle that differs by a term of the form $\prod_{p_i \in S} p_i^{-v_i}$ from the one described in [15].

Our input can be split between $(u_1, \dots, u_{n_1+n_2}, v_1, \dots, v_{|S|}) \in \mathbb{R}^{n_1+n_2} \times \mathbb{Z}^{|S|}$ and a phase $\theta \in \mathbb{Z}_2^{n_1} \times (\mathbb{R}/\mathbb{Z})^{n_2}$. As mentioned in [15], the phase can be dealt with separately and is not computationally problematic. To make our presentation simpler, we show how to split up the computation in the absence of phase. To avoid the expensive computations with the e^{u_i} , we use E -ideal arithmetic which we analyze in Section 4.2. Our main concern when splitting up the computation is that we want to reduce it to operations between E -ideals of determinant $\sqrt{|\Delta|}$. This gives us upper and lower bounds on the vectors in play, which in turns bounds the computational complexity of arithmetic operations as we see in Section 4.3.

Let $(u_1, \dots, u_{n_1+n_2}, v_1, \dots, v_{|S|}) \in \mathbb{R}^{n_1+n_2} \times \mathbb{Z}^{|S|}$ be an input vector where u_1 satisfies the condition given by (1). We can separate the evaluation of the oracle in two steps by rewriting it as

$$(u_1, \dots, u_{n_1+n_2-1}, u'_{n_1+n_2}, 0, \dots, 0) + \left(0, \dots, 0, \frac{1}{2} \sum_j e_j v_j \log(p_j), v_1, \dots, v_{|S|} \right).$$

where $u'_{n_1+n_2} = -\frac{1}{2} \sum_{j \leq n_1} u_j - \sum_{n_1 < j < n_1+n_2} u_j$. The first term is evaluated the same way as [15]. More specifically, we separate real numbers between integer and fractional part. We define $(r_j)_{j \leq n_1+n_2} \in \mathbb{Z}^{n_1+n_2}$ and $(s_j)_{j \leq n_1+n_2} \in [0, 1)^{n_1+n_2}$ by $u_j = r_j + s_j$ for $j < n_1 + n_2$, $r_{n_1+n_2} := -\sum_{j < r_1+r_2} r_j$ and $s_{n_1+n_2} := u'_{n_1+n_2} - r_{n_1+n_2}$. As $s_i < 1$, we calculate e^{s_i} to a given precision q by using the formula $e^x = \sum_{k \leq M} \frac{x^k}{k!} + O(x^{M+1})$. The number of terms in the sum has to satisfy $M \in O(q)$. This way, we can compute $\phi(s_1, \dots, s_{n_1+n_2}) = (e^{s_1}, \dots, e^{s_{n_1+n_2}})$ and the corresponding E -ideal $A_{-1} := (e^{s_1}, \dots, e^{s_{n_1+n_2}}) \cdot \underline{\mathcal{Q}}$ by multiplication with each generator of $\underline{\mathcal{Q}}$. Let $(a_k^{(j)}) \in \{-1, 0, 1\}$ be such that $r_j = \sum_{k \leq \lceil \log_2(r_j) \rceil} a_k^{(j)} 2^k$ is the binary decomposition of r_j for $j < n_1 + n_2$ and $a_k^{(n_1+n_2)} := -\sum_{j < n_1+n_2} a_k^{(j)}$ and $\log_2(r) := \max_j \lceil \log_2(r_j) \rceil$. Note that we have $u'_{n_1+n_2} = \sum_k a_k^{(n_1+n_2)} 2^k$, but the $a_k^{(n_1+n_2)}$ are not its binary decomposition. They take values in $[-n_1 - n_2, n_1 + n_2]$. The E -ideal generated by the integer part of the u_i satisfies

$$\begin{aligned} (e^{r_1}, \dots, e^{r_{n_1+n_2}}) \cdot \underline{\mathcal{Q}} &= \prod_{k \leq \log_2(r)} \left(e^{a_1^{(k)} 2^k}, \dots, e^{a_{n_1+n_2}^{(k)} 2^k} \right) \cdot \underline{\mathcal{Q}} \\ &= \prod_{k \leq \log_2(r)} \left[\underbrace{\left(e^{a_1^{(k)}}, \dots, e^{a_{n_1+n_2}^{(k)}} \right) \cdot \underline{\mathcal{Q}}}_{A_k} \right]^{2^k}. \end{aligned} \quad (2)$$

The norm of the E -ideals A_k for $k \leq \log_2(r)$ is $\mathcal{N}(A_k) = e^{\sum_j a_j^{(k)}} \mathcal{N}(\underline{\mathcal{Q}}) = 1$. Therefore $\det(A_k) = \sqrt{|\Delta|}$.

Likewise, the bit size of $e^{e_1 v_i \log(p_i)}$ is at least proportional to v_i , and therefore exponential in the bit size of v_i which is part of the input. Therefore, we need to split up the computation

of the E -ideal

$$\left(0, \dots, 0, \frac{1}{2} \sum_j e_j v_j \log(p_j), v_1, \dots, v_{|S|}\right) \mapsto \left(1, \dots, 1, e^{\frac{1}{2} \sum_j e_j v_j \log(p_j)}\right) \cdot \underline{\mathcal{O}} \cdot \prod_j \mathfrak{p}_j^{-v_j}.$$

Let $(b_j^{(k)})$ such that $v_j = \sum_{k \leq \lceil \log_2(v_j) \rceil} b_j^{(k)} 2^k$ and $\log_2(v) := \max_j \log_2(v_j)$. Then we have the decomposition

$$\begin{aligned} \left(1, \dots, 1, e^{\frac{1}{2} \sum_j e_j v_j \log(p_j)}\right) \cdot \underline{\mathcal{O}} \cdot \prod_j \mathfrak{p}_j^{-v_j} &= \prod_{j \leq |S|} \left(\left(1, \dots, 1, e^{e_j \log(p_j)}\right) \cdot \underline{\mathcal{O}} \cdot \mathfrak{p}_j^{-1} \right)^{v_j} \\ &= \prod_{j \leq |S|} \prod_{k \leq \log_2(v)} \left(\left(1, \dots, 1, e^{e_j \log(p_j)}\right) \cdot \underline{\mathcal{O}} \cdot \mathfrak{p}_j^{-1} \right)^{b_j^{(k)} 2^k} \\ &= \prod_{k \leq \log_2(v)} \left(\prod_{j \leq |S|} \left(\underbrace{(1, \dots, 1, p^{e_j}) \cdot \underline{\mathcal{O}} \cdot \mathfrak{p}_j^{-1}}_{B_{j,k}} \right)^{b_j^{(k)} 2^k} \right) \end{aligned} \quad (3)$$

The calculation is decomposed the following way: first compute $B_k := \prod_{j \leq |S|} B_{j,k}^{b_j^{(k)}}$ which involves $\log_2(v) \cdot |S|$ multiplications between the E -ideals $B_{j,k}$ which have determinant $\sqrt{|\Delta|}$, and then return $\prod_{k \leq \log_2(v)} B_k^{2^k}$ which requires at most $\log_2(v)^2$ multiplications between the E -ideals B_k which also have determinant $\sqrt{|\Delta|}$.

Algorithm 1 Classical oracle evaluation (without phase)

Input: $(u_2, \dots, u_{n_1+n_2}, v_1, \dots, v_{|S|})$.

Output: The E -ideal corresponding to $\phi(u_1, \dots, u'_{n_1+n_2}) \cdot \underline{\mathcal{O}} \prod_j \mathfrak{p}_j^{-v_j}$.

- 1: Compute u_1 according to (1)
 - 2: Compute A_{-1} using the formula $e^x \simeq \sum \frac{x^i}{i!}$.
 - 3: Compute the A_j using (2).
 - 4: Compute the $B_{j,k}$ using (3).
 - 5: For each $k \leq \log_2(v)$, $B_k \leftarrow \prod_j B_{j,k}$.
 - 6: **return** $A_{-1} \cdot \prod_j A_j^{2^j} \cdot \prod_k B_k^{2^k}$.
-

Proposition 1. *Algorithm 1 is correct and involves a polynomial number of multiplications between E -ideals of determinant $\sqrt{|\Delta|}$.*

4.2 E -ideal arithmetic

The arithmetic between E -ideals is directly inspired from the arithmetic between ideals in a number field. To evaluate our classical oracle, we need an efficient implementation of the E -ideal multiplication. Let $A = \oplus_{j \leq n} \mathbb{Z}a_j$ and $B = \oplus_{k \leq n} \mathbb{Z}b_k$ be E -ideals generated by the

$a_j, b_k \in E$. Then the E -ideal $A \cdot B$ is the lattice generated by the n^2 elements $(a_j \cdot b_k)_{j,k \leq n}$. The multiplication of two E -ideals can be described by the two following steps:

1. Calculate all the cross terms $a_j \cdot b_k$ for $j, k \leq n$.
2. Compute a basis $(c_j)_{j \leq n}$ of $\sum_{j,k} \mathbb{Z} a_j \cdot b_k$.

The main challenge of E -ideal multiplication is that we need to deal with rational approximations of lattices. We need to estimate how much precision is needed to ensure accuracy, and how much precision is lost after each operation. We employ the same strategy as in [15], which uses fixed point rational approximations such that $\left| \frac{\hat{a}}{2^{q_0}} - a \right| \leq \frac{1}{2^q}$ where the precision q deteriorates throughout the computation.

Multiplication of approximate elements E -ideals can be seen as lattices in $\mathbb{R}^m$ where $m = n_1 + 4n_2$ by mapping elements in E of the form $(u_1, \dots, u_{n_1+n_2})$ to

$$(u_1, \dots, u_{n_1}, \Re(u_{n_1+1}), \Im(u_{n_1+1}), \Re(u_{n_1+1}), -\Im(u_{n_1+1}), \dots, u_m) \in \mathbb{R}^m. \quad (4)$$

We approximate each coordinate $a \in \mathbb{R}$ of such vector by $\hat{a}/2^{q_0}$ with $\hat{a} \in \mathbb{Z}$ and $q_0 \in \mathbb{Z}_{>0}$ satisfying $\left| \frac{\hat{a}}{2^{q_0}} - a \right| \leq \frac{1}{2^q}$ for some $q \leq q_0 + 1$. Then the multiplication of q -bit approximations of $a, b \in \mathbb{R}$ with $\log_2(|a|), \log_2(|b|) \leq c$ satisfies

$$\left| \frac{\hat{a}\hat{b}}{2^{2q_0}} - ab \right| = \left| \left(\frac{\hat{a}}{2^{q_0}} - a \right) \frac{\hat{b}}{2^{q_0}} + \left(\frac{\hat{b}}{2^{q_0}} - b \right) a \right| \leq \frac{2^c}{2^q} + \frac{2^c}{2^q} = \frac{1}{2^{q-c-1}}.$$

This gives us a rational number approximating ab , but the denominator increases, which reduces the efficiency of the computation of a basis of the E -ideal generated by the n^2 products. Let $\hat{ab} := \left\lfloor 2^{q_0} \left(\frac{\hat{a}\hat{b}}{2^{2q_0}} \right) \right\rfloor$, then

$$\left| \frac{\hat{ab}}{2^{q_0}} - ab \right| \leq \left| \frac{\hat{ab}}{2^{q_0}} - \frac{\hat{a}\hat{b}}{2^{2q_0}} \right| + \left| \frac{\hat{a}\hat{b}}{2^{2q_0}} - ab \right| \leq \frac{1}{2^{q_0+1}} + \frac{1}{2^{q-c-1}} \leq \frac{1}{2^{q-c-1}} + \frac{1}{2^{q-c-1}} = \frac{1}{2^{q-c-2}}.$$

This gives us the approximation of the multiplication of two real numbers. Although we approximate vectors in E by vectors in $\mathbb{R}^m$, the pointwise multiplication of the $m - n_1$ last coordinates correspond to multiplications between complex numbers. The addition involved in the complex number multiplication $(a + ib)(c + id) = (ac - bd) + i(ad + bc)$ induces the loss of an extra bit of precision. If $a, b \in \mathbb{R}$ are approximated by $\hat{a}, \hat{b}$ with precision q , then $\widehat{a+b} := (\hat{a} + \hat{b})$ satisfies $\left| \frac{\widehat{a+b}}{2^{q_0}} - (a+b) \right| \leq \frac{1}{2^{q-1}}$. We summarize pointwise multiplication of approximations of elements in Algorithm

Lemma 2. *Algorithm 2 returns a vector $\hat{ab} \in \mathbb{Z}^m$ whose entries approximates those of $ab \in E$ with precision $q - c - 3$ in time $\tilde{O}(m(q_0 + c))$. Additionally, this vector satisfies*

$$\left\| \frac{\hat{ab}}{2^{q_0}} - ab \right\| \leq \frac{\sqrt{m}}{2^{q-c-3}}$$

Algorithm 2 Multiplication between approximations of elements in E

Input: $\hat{a}, \hat{b} \in \mathbb{Z}^m$ such that $\left| \frac{\hat{a}_j}{2^{q_0}} - a_j \right| \leq \frac{1}{2^q}$ and $\left| \frac{\hat{b}_j}{2^{q_0}} - b_j \right| \leq \frac{1}{2^q}$ where $(a_j), (b_j)$ are the entries of $a, b \in E$ and $c \geq \log_2(\|a\|), \log_2(\|b\|)$.

Output: An approximation $\hat{ab} \in \mathbb{Z}^m$ of the vector $ab \in \mathbb{R}^m$ with precision $q' := q - c - 3$.

- 1: **For** $j \leq n_1$ **do** $\hat{ab}_j := \left\lfloor 2^{q_0} \left(\frac{\hat{a}_j \hat{b}_j}{2^{2q_0}} \right) \right\rfloor$.
 - 2: **while** $n_1 < j \leq m$ **do**
 - 3: $w := \left\lfloor 2^{q_0} \left(\frac{\hat{a}_j \hat{b}_j}{2^{2q_0}} \right) \right\rfloor$, $x := \left\lfloor 2^{q_0} \left(\frac{\hat{a}_{j+1} \hat{b}_{j+1}}{2^{2q_0}} \right) \right\rfloor$.
 - 4: $y := \left\lfloor 2^{q_0} \left(\frac{\hat{a}_j \hat{b}_{j+1}}{2^{2q_0}} \right) \right\rfloor$, $z := \left\lfloor 2^{q_0} \left(\frac{\hat{a}_{j+1} \hat{b}_j}{2^{2q_0}} \right) \right\rfloor$.
 - 5: $\hat{ab}_j := w - x$, $\hat{ab}_{j+1} := y + z$, $j \leftarrow j + 2$.
 - 6: **end while**
 - 7: **return** $(\hat{ab}_j)$.
-

Computing a basis from a generating set Let Λ be an E -ideal for which we want to find a basis of short vectors in polynomial time. As the Euclidean norm is preserved by the mapping of Λ in $\mathbb{R}^m$, this problem boils down to computing a short basis of an ideal in $\mathbb{R}^m$. Since the original description of the LLL reduction algorithm [25], the problem of finding a short basis (up to an approximation factor) of a lattice in polynomial time is well understood. The difficulty in this context is that we are dealing with rational approximations of real numbers. Let $a = (a_1, \dots, a_m) \in \mathbb{R}^m$ and $q_0 + 1 \geq q \geq 0$, we say that $\hat{a} = (\hat{a}_1, \dots, \hat{a}_m) \in \mathbb{Z}^m$ is an approximation of a with precision q if $\forall j \leq m$, $\left| \frac{\hat{a}_j}{2^{q_0}} - a_j \right| \leq \frac{1}{2^q}$. Given an approximate generating set for the lattice $\Lambda \in \mathbb{R}^m$, we want to compute a basis of short vectors that approximates a basis of short vectors for $\Lambda \in \mathbb{R}^m$. We rely on a result from Buchmann and Kessler [7] and its modification by Eisenträger, Halgren, Kitaev and Song [15].

Let $\hat{a}_1, \dots, \hat{a}_k \in \mathbb{Z}^m$ be rational approximations of $a_1, \dots, a_k \in \mathbb{R}^m$ of precision q (and denominator $q_0 \geq q - 1$). Let $r \leq k$ be the rank of the lattice generated by $(a_j)_{j \leq k}$. The approach described in [7] consists of applying the LLL reduction algorithm to the rank k lattice generated by the independent vectors $\tilde{a}_j := (e_j, \hat{a}_j)$, $j \leq k$ where e_j is the j -th unit vector of $\mathbb{Z}^k$. The LLL algorithm outputs vectors $\tilde{b}_j = (m_j, \hat{b}_j)$, $j \leq k$ such that if the input precision q is large enough, $m_1, \dots, m_{k-r}$ are independent relations for $a_1, \dots, a_k$ (i.e. $\sum_l m_l^{(j)} a_j^{(l)} = 0$) and the vectors $b_j = \sum_j m_{k-r+j}^{(l)} a_l$, $j \leq r$ are a basis for the lattice $\sum_j \mathbb{Z} a_j$.

The following proposition states our modification of the result of [7] incorporating the cases where $q_0 \geq q$.

Proposition 2 (Theorem C.5 of [16]). *Let $a_1, \dots, a_k \in \mathbb{R}^m$ be a generating set for a lattice L of rank r . Let $\hat{a}_1, \dots, \hat{a}_k \in \mathbb{Z}^m$ be rational approximations of the a_i with precision q and denominator q_0 (i.e. $\forall i \leq k$, $\|\hat{a}_i/2^{q_0} - a_i\| \leq \sqrt{m}/2^{2^q}$). Assume μ is a lower bound on $\lambda_1(L)$, and that*

$$2^q \geq \left(k 2^{\frac{k+1}{2}} \max \|a_i\| \right)^r / (\mu \det(L)^2).$$

Then Algorithm 3 returns approximations $\hat{c}_1, \dots, \hat{c}_r \in \mathbb{Z}^r$ of a basis $c_1, \dots, c_r$ of L that

Algorithm 3 Computation of a basis from an approximate generating set

Input: Approximate vectors $(\hat{a}_j)_{j \leq k} \in \mathbb{Z}^m$ of $(a_j)_{j \leq k} \in \mathbb{R}^m$ of precision q and denominator q_0 .

Output: Approximate vectors $(\hat{b}_j)_{j \leq r} \in \mathbb{Z}^m$ of a basis of $\sum_j \mathbb{Z}a_j$.

- 1: **for** $j \leq k$ **do** $\tilde{a}_j \leftarrow (e_j, \hat{a}_j)$.
 - 2: Apply LLL to the $\tilde{a}_j$ and get $\tilde{b}'_j := (m'_j, \hat{b}'_j)$.
 - 3: Apply LLL to the $\hat{b}'_j$ for $k - r + 1 \leq j \leq k$. and get $\hat{b}_{j-k+r}$ for $k - r + 1 \leq j \leq k$.
 - 4: **return** $\hat{b}_l$ for $l \leq r$.
-

satisfies

$$\forall i \leq r, \|c_i\| \leq (\sqrt{mk} + 2) 2^{\frac{k-1}{2}} \lambda_i(L).$$

Additionally, the precision of the output satisfies $\|\hat{c}_i/2^{q_0} - c_i\| \leq rk\gamma_1\gamma_3\sqrt{m}/2^q$ where

- $\gamma_1 \leq k^2 2^{\frac{k+1}{2}} \frac{\max_i \|a_i\|}{\det(L)}$
- $\gamma_3 \leq \frac{4k(k^{2^{k/2}} \max_i \|a_i\|)^r}{\det(L)^2}$.

The cost of this procedure is

$$\text{Cost}_{LLL}(k, k, \log_2(\max_i \|a_i\|) + q_0) + \text{Cost}_{LLL}\left(r, k, \log_2\left(k(\sqrt{km} + 2) 2^{\frac{k-1}{2}} \max_i \|a_i\|\right) + q_0\right),$$

where $\text{Cost}_{LLL}(r, d, b)$ denotes the cost of running the LLL algorithm on r linearly independent vectors of $\mathbb{Z}^d$ with entries of bit size bounded by b .

Proof. The bounds on the size of the entries and on the precision of the output are exactly the statement made in [16, Th. C.5]. To derive the bound on the cost of the procedure, one needs to use information in the proof of [16, Th. C.5] regarding the size of the entries of the vectors given as input to the second call to LLL. Indeed, it is proved that the vectors have length bounded by $k(\sqrt{km} + 2) 2^{\frac{k-1}{2}} \max_i \|a_i\|$. $\square$

4.3 Complexity of the classical computation

To estimate the asymptotic complexity of the classical oracle, we need to combine the results of Section 4.1 and Section 4.2. Let $(y, v_1, \dots, v_{|S|}) \in \mathbb{R}^{n_1+n_2-1} \times \mathbb{Z}_2^{n_1} \times (\mathbb{R}/\mathbb{Z})^{n_2} \times \mathbb{Z}^{|S|}$. We want to compute a poly-size basis of

$$f_c(y, v_1, \dots, v_{|S|}) = \phi(y) \cdot \underline{\mathcal{O}} \cdot \mathbf{p}_1^{-v_1} \dots \mathbf{p}_{|S|}^{-v_{|S|}},$$

in polynomial time in $\max_j \{\log(|y_j|)\}$, $|S|$, $\max_j \{\log(p_j)\}$, $\max_j \{\log |v_j|\}$, m , and $\log |\Delta|$.

Theorem 2. *Let K be a number field of degree n and discriminant Δ . Let S be a set of prime ideals of K . The gate complexity of the classical oracle is in*

$$\mathcal{O}\left((\log_2(v)(|S| + \log_2(v)) + \log_2(r)^2) C_{\text{gates}}(n^2, \beta)\right),$$

and the qubit requirement is in

$$O\left(\max(|S|, \log_2(v), \log_2(r)) C_{\text{qubits}}(n^2, \beta)\right)$$

where $v = \max_i |v_i|$, $r = \max_i |y_i|$,

$$\beta \in \tilde{O}\left(n(n^2 + \log |\Delta|) \cdot \max(\log |S| + \log(v), \log(r))\right),$$

is a bound on the bit size of the integer vectors given to the LLL algorithm, and the costs $C_{\text{gates}}(k, b)$, $C_{\text{qubits}}(k, b)$ denote the gate cost (resp. memory cost) of the LLL algorithm on input an k -rank lattice in $\mathbb{Z}^k$ with entries of bit size bounded by b . As usual $\tilde{O}$ denotes the complexity where the logarithmic factors are omitted.

Proof. Given how we split up the computation, we only multiply E -ideals of determinant $\sqrt{|\Delta|}$. This means that once the original precision q_0 is chosen, the cost of each ideal multiplication is bounded. Let C_{gates} be a bound on the number of quantum gates required for the multiplication of two ideals, and C_{qubits} be a bound on the amount of memory required to multiply two ideals. On input $(u, v) \in \mathbb{R}^{n_1+n_2} \times \mathbb{Z}^{|S|}$ the evaluation of f_c consists in the calculation of $\prod_{k \leq \log_2(r)} A_k^{2^k}$ and $\prod_{k \leq \log_2(v)} B_k^{2^k}$ where $\log_2(r) \leq \max_j (\log_2(|u_j|))$, $\log_2(v) \leq \max_j (v_j)$, and the A_k, B_k are defined in Section 4.1.

We first analyze the cost of computing $\prod_{k \leq \log_2(v)} B_k^{2^k}$. For each $k \leq \log_2(v)$, we compute $B_k := \prod_{j \leq |S|} B_{j,k}^{b_j^{(k)}}$. To minimize the length of the multiplication chain, we use a multiplication tree of depth $\log_2(|S|)$. This costs $O(|S|C_{\text{gates}})$ gates and uses $O(|S|C_{\text{qubits}})$ qubits. Then we compute $B_k^{2^k}$ by performing k squarings with $O(kC_{\text{gates}})$ gates and $O(kC_{\text{qubits}})$ qubits. This procedure is repeated for all $k \leq \log_2(v)$ using a total of $O(\log_2(v)(|S| + \log_2(v))C_{\text{gates}})$ gates and $O(\max(|S|, \log_2(v))C_{\text{qubits}})$ qubits. Finally, we compute the product $\prod_{k \leq \log_2(v)} B_k^{2^k}$ via a product tree of depth $\log_2 \log_2(v)$ using $O(\log_2(v)C_{\text{gates}})$ gates and $O(\log_2(v)C_{\text{qubits}})$ qubits. In summary, $\prod_{k \leq \log_2(v)} B_k^{2^k}$ is computed using $O(\log_2(v)(|S| + \log_2(v))C_{\text{gates}})$ gates and $O(\max(|S|, \log_2(v))C_{\text{qubits}})$ qubits. The longest multiplication chain leading to this value is of length $\log_2(|S|) + \log_2(v) + \log_2 \log_2(v)$.

Next, we analyze the cost of computing $\prod_{k \leq \log_2(r)} A_k^{2^k}$. We follow the same strategy as above. First we compute all the $A_k^{2^k}$ for $k \leq \log_2(r)$ using $O(\log_2(r)^2 C_{\text{gates}})$ gates and $O(\log_2(r)C_{\text{qubits}})$ qubits. The product $\prod_{k \leq \log_2(r)} A_k^{2^k}$ is then computed with a product tree of depth $\log_2 \log_2(r)$ using an extra $O(\log_2(r)C_{\text{gates}})$ gates. The total number of gates required is in $O(\log_2(r)^2 C_{\text{gates}})$, and the number of required qubits in $O(\log_2(r)C_{\text{qubits}})$. The longest multiplication chain is of length $\log_2(r) + \log_2 \log_2(r)$.

In summary, the gate cost, memory cost and longest multiplication chains of the computation satisfy:

- Gate cost in $O\left((\log_2(v)(|S| + \log_2(v)) + \log_2(r)^2) C_{\text{gates}}\right)$.
- Memory cost in $O\left(\max(|S|, \log_2(v), \log_2(r)) C_{\text{qubits}}\right)$.
- Longest multiplication chain: $\max(\log_2 |S| + \log_2(v) + \log_2 \log_2(v), \log_2(r) + \log_2 \log_2(r))$.

To determine the cost of each multiplication, we need to evaluate the size of the entries of the matrices that are given as input to the LLL algorithm. This depends on the precision we require. For Proposition 2 to apply, we need that the final precision q satisfy $2^q \geq \left(k2^{\frac{k+1}{2}} \max \|a_i\|\right)^r / (\mu \det(L)^2)$ where μ is a lower bound on $\lambda_1(L)$. We have that $\det(L) = \sqrt{|\Delta|}$ and $\lambda_1(L) \geq \sqrt{n}$. Moreover, we can always assume that matrix given as input is reduced, which means that $\|a_i\| \leq (\sqrt{nk} + 2) 2^{\frac{k-1}{2}} \lambda_i(L) \leq (\sqrt{nk} + 2) 2^{\frac{k-1}{2}} \sqrt{n|\Delta|}$. At each ideal multiplication, the number q_{loss} of bits of precision we lose satisfies

$$q_{\text{loss}} \leq \log_2 \left((\sqrt{nk} + 2) 2^{\frac{k-1}{2}} \sqrt{n|\Delta|} \right) + \log_2 (n^{3.5} \gamma_1 \gamma_3),$$

where γ_1, γ_3 are defined in Proposition 2. If we define $\ell_{\text{mult}} := \max(\log_2 |S| + \log_2(v) + \log_2 \log_2(v), \log_2(r) + \log_2 \log_2(r))$ to be the length of the longest multiplication chain, and the target final precision $q_{\text{final}} := n \log_2 \left(k \sqrt{n} (\sqrt{nk} + 2) 2^k \sqrt{|\Delta|} \right)$, then the initial precision is $q_0 := q_{\text{final}} + \ell_{\text{mult}} q_{\text{loss}}$. To simplify the asymptotic estimate of the cost of the multiplication of ideals, we notice that the bound α on the bit size of the vectors of the basis of L given as input to the first LLL solver, and the bound α' on the bit size of the basis vectors given to the second LLL solver both satisfy:

$$\alpha, \alpha' \in \tilde{O}(n^2 + \log_2 |\Delta|).$$

This means that the cost of each multiplication is dominated by the cost of LLL on input a rank and dimension n^2 lattice with vectors of bit size in $\tilde{O}(n^2 + \log_2 |\Delta| + q_0)$. We then notice that

$$q_0 \in \tilde{O}(n(n^2 + \log |\Delta|) + \max(\log |S| + \log(v), \log(r))),$$

which asymptotically dominates the bit size of the integer vectors given as input to the LLL algorithm. □

4.4 The quantum encoding of $e^t \mathcal{Q} \prod_{\mathfrak{p} \in S} \mathfrak{p}^{-v_i}$

Let $g_s(\cdot)$ be the Gaussian function $g_s(x) := e^{-\pi \|x\|^2 / s^2}$, $x \in \mathbb{R}^n$. For any set $S \subset \mathbb{R}^n$, denote $g_s(S) := \sum_{x \in S} g_s(x)$. Given a lattice L , the quantum encoding maps L to the lattice Gaussian state via

$$\begin{aligned} \{\text{Lattices over } E\} & \xrightarrow{f_q} \mathcal{S} \text{ (unit vectors in a Hilbert space)} \\ L & \longrightarrow |L\rangle := \gamma \sum_{v \in L} g_s(v) |\text{str}_{\nu,n}(v)\rangle \end{aligned}$$

where γ is a factor that normalized the state. Here $|\text{str}_{\nu,n}(v)\rangle$ is the straddle encoding of a real-valued vector $v \in \mathbb{R}^n$, as defined in [15]. Intuitively, we discretize the space $\mathbb{R}^n$ by a grid $\nu \mathbb{Z}^n$, and we encode the information about v by a superposition over all grid nodes surrounding v . Specifically, for the one-dimensional case, the straddle encoding of a real number is

$$x \in \mathbb{R} \mapsto |\text{str}_{\nu}(x)\rangle := \cos\left(\frac{\pi}{2}t\right)|k\rangle + \sin\left(\frac{\pi}{2}t\right)|k+1\rangle,$$

where $k := \lfloor x/\nu \rfloor$ denotes the nearest grid point no bigger than x and $t := x/\nu - k$ denotes the (scaled) offset. Repeat this for each coordinate of $v = (v_1, \dots, v_n)$ we get $|\text{str}_{\nu,n}(v)\rangle := \bigotimes_{i=1}^n |\text{str}_{\nu}(v_i)\rangle$. We recall some properties about straddle encoding from [15]. This will be useful to prove the HSP properties of our function.

Fact 1. *Let $v, w \in \mathbb{R}^n$. The following hold*

- $\| |\text{str}_{\nu,n}(v)\rangle - |\text{str}_{\nu,n}(w)\rangle \| \leq \frac{\pi}{2\nu} \sqrt{n} \cdot \|v - w\|$.
- *If $\|v - w\| \geq 2\sqrt{n}\nu$, then $\langle \text{str}_{\nu,n}(v) | \text{str}_{\nu,n}(w) \rangle = 0$.*

In our lattice Gaussian states, we will always make sure $\lambda_1(L) > 2\sqrt{n}\nu$ so that

$$\langle \text{str}_{\nu,n}(v) | \text{str}_{\nu,n}(u) \rangle = 0 \text{ whenever } v \neq u.$$

In this case we can compute the normalization factor $\gamma = \left(g_{\frac{s}{\sqrt{2}}}(L)\right)^{-1/2}$. As shown in [15], one can efficiently compute f_q if the lattice satisfies certain conditions and a good basis is given (e.g., L is LLL-reduced). Namely there is an efficient quantum circuit creating lattice Gaussian states. We state this result with gate and qubit count estimation below and will invoke it as a black-box.

Fact 2. *Let L be an LLL-reduced basis. Assume that $\lambda_1(L) \geq \lambda_0$, $\det(L) \leq d_0$ and $s \geq n^{n/2+1} 2^{3n} \lambda_0^{-n+1} d_0$. Let $|L\rangle = \gamma \sum_{v \in L} g_s(v) |\text{str}_{\nu,n}(v)\rangle$. There is a quantum circuit that takes L as input and produces a state $|\psi\rangle$ such that $\| |\psi\rangle - |L\rangle \| \leq 2^{-n}$. The quantum circuit has gate count $O(n^3 + n \log(1/\nu))$ and qubit count $O(n \log(\sqrt{n} \|L^{-1}\| s)) = O(n^2(\log n + \log(1/\lambda_0)))$.*

5 Pseudoinjectivity of $(\mathbf{y}, \mathbf{v}) \mapsto |\phi(\mathbf{y})\mathcal{Q} \prod_{\mathbf{p} \in S} \mathfrak{p}^{-v_i}\rangle$

Theorem 3. *Let f be the function $G = \mathbb{R}^{n_1+n_2-1} \times \mathbb{Z}_2^{n_1} \times (\mathbb{R}/\mathbb{Z})^{n_2} \times \mathbb{Z}^s \rightarrow \mathcal{H}$ defined by $(\mathbf{y}, \mathbf{v}) \mapsto |\phi(\mathbf{y})\mathcal{Q} \prod_{\mathbf{p} \in S} \mathfrak{p}^{-v_i}\rangle$. There is $r, \varepsilon > 0$ such that*

$$d_{G/L}(x, y) := \min_{v \in U_S^G} \|x - y - v\| \geq r \Rightarrow |\langle f(x) | f(y) \rangle| \leq \varepsilon$$

Our proof relies on some statements on lattices available in [16]. As in [16, Sec. E.2], we first introduce a central notion called the *approximate intersecting sublattice* of two lattices L and L' in $\mathbb{R}^m$.

Definition 5 (δ -approximate intersecting sublattice). *Let L and L' be two lattices of dimension n in $\mathbb{R}^m$. Let $Y := \{(x, x') : x \in L_R, x' \in L'_R, \|x - x'\| \leq \delta\}$ and $X := Y|_1$ ($X' := Y|_2$) be the corresponding set of points x (resp. x'). Define $\Lambda := \langle X \rangle$ ($\Lambda' := \langle X' \rangle$ resp.) be the sublattice generated by points in X (X' resp.). We call Λ (Λ') the δ -approximate intersecting sublattice of L (resp. L') between L and L' .*

Here $L_R = L \cap \mathbf{B}_R$ are the lattice points inside a sphere of radius $R = \sqrt{ns}$, where s is the Gaussian width in the lattice Gaussian state. This definition indeed captures the overlap (up to δ -approximation) between two lattices. Intuitively, Λ and Λ' can be paired up that

are “close”, and all the other pairs of points will be “far” apart. This overlap is the main contribution to the inner product between the quantum encoding of two lattices, and we show that if it generates a proper sublattice, we can bound the scalar product. This is formalized below as shown in [15]. We sketch a proof for completeness.

Fact 3 (Lemma E.6 of [16]). *Let L, L', Λ and Λ' be as in Definition 5. Suppose that: $\lambda_1 \geq \lambda$, $\lambda'_1 \geq \lambda$. Then there is a one-one correspondence $h : \Lambda \rightarrow \Lambda'$ such that*

- $\forall x \in \Lambda, \|x - h(x)\| \leq \beta \|x\|$ with $\beta := n(\sqrt{n}R/\lambda)^n \cdot \frac{\delta}{R}$;
- For any $x \in L_R$ and any $x' \in L'_R$, if $x' \neq h(x)$ (in particular if $x \notin \Lambda$ or $x' \notin \Lambda'$), $\|x - x'\| > \delta$.

Proof. (Sketch) Pick $x_i \in X : i = 1, \dots, n$ that are linearly independent and let x'_i be the corresponding points in X' . Let $h : x_i \mapsto x'_i$ and this extends to a linear map from $\bar{\Lambda}$ to $\bar{\Lambda}'$. The second property holds immediately by definition. To show the first one, let $x \in \Lambda$ and write it as $x = \sum_i \alpha_i x_i, \alpha_i \in \mathbb{R}$. Using Cramer’s rule, Hadamard inequality and Minkowski’s second theorem, one can get $|\alpha_i| \leq (\sqrt{n}R/\lambda)^n \frac{\|x\|}{R}$. Therefore $\|x - h(x)\| = \|\sum_i \alpha_i (x_i - h(x_i))\| \leq \sum_i |\alpha_i| \delta \leq \beta \|x\|$ with $\beta = n(\sqrt{n}R/\lambda)^n \cdot \frac{\delta}{R}$. $\square$

If we pick the straddle encoding fine enough such that $2\sqrt{n}\nu < \delta$, it follows that the inner product between their quantum encodings will be solely contributed by Λ and Λ' . In particular:

Fact 4 (Lemma E.7 of [16]). *Let $\mathcal{L}$ and $\mathcal{L}'$ be two E -ideals with $\max\{\det(\mathcal{L}), \det(\mathcal{L}')\} \leq d$ and $\min \lambda_1(\mathcal{L}), \lambda_1(\mathcal{L}') \geq \lambda$. Let Λ and Λ' be the δ -intersecting sublattices of $\mathcal{L}$ and $\mathcal{L}'$ respectively, as defined in Definition 5. If $\Lambda \subsetneq \mathcal{L}$ (which implies $\Lambda' \subsetneq \mathcal{L}'$), then $\langle \mathcal{L} | \mathcal{L}' \rangle \leq 3/4$ whenever $s \geq 4\pi n^{n/2+3}d/\lambda^{n-1}$.*

The two previous claims give us a sufficient condition for $\langle \mathcal{L} | \mathcal{L}' \rangle \leq 3/4$. To prove the (r, ε) -condition, we need to relate the properties of Λ to our notion of distance between the preimages in G . We first prove a sufficient condition on $\text{dist}(\mathcal{L}, \mathcal{L}')$ in Lemma 3, which ensures that the approximate intersecting sublattices Λ and Λ' be proper.

Lemma 3. *If $\text{dist}(\mathcal{L}, \mathcal{L}') \geq r = \frac{1}{2\sqrt{n}|\Delta|}$ and $\beta < \frac{1}{20n^{n+2}|\Delta|}$, then the δ -intersecting sublattices Λ and Λ' of $\mathcal{L}$ and $\mathcal{L}'$ respectively, as defined in Definition 5, become proper sublattices. Namely $\Lambda \subsetneq \mathcal{L}$ and $\Lambda' \subsetneq \mathcal{L}'$.*

On the other hand, if $\Lambda = \mathcal{L}$ and $\Lambda' = \mathcal{L}'$, then there is W satisfying $M_{\mathcal{L}'} = M_{\mathcal{L}}W$ for any bases $M_{\mathcal{L}'}, M_{\mathcal{L}}$ of $\mathcal{L}', \mathcal{L}$ that is of the form $W = e^{\text{diag}(a)}$ for some a with $\|a\| \leq \frac{1}{4\sqrt{n}|\Delta|}$.

Proof. Suppose for contradiction that $\Lambda = \mathcal{L}$ and $\Lambda' = \mathcal{L}'$. Let M_h be the matrix induced by h (wrt to some choice of basis for $\mathcal{L}$ and $\mathcal{L}'$). First we claim that $\|M_h - I\|_\infty \leq \beta^{(1)} := n^{n+1}\beta$. To show this, we pick a short basis $(v_1, \dots, v_n)$ for $\mathcal{L}$ such that $\|v_k\| \leq \sqrt{k}\lambda_k(\mathcal{L})$ for $k \leq n$, which always exists. Then any $w \in \mathbb{R}^n$ with $\|w\| = 1$ can be written as $w = \sum_i \alpha_i v_i, \alpha_i \in \mathbb{R}$. By Cramer’s rule we have

$$|\alpha_i| = \left| \frac{\det(v_1, \dots, v_{i-1}, w, v_{i+1}, \dots, v_n)}{\det(v_1, \dots, v_n)} \right| \leq \frac{(\sqrt{n})^n \prod_{j \neq i} \lambda_j(\mathcal{L})}{\sqrt{i} \det(\mathcal{L})} \leq n^n / \sqrt{i} \lambda_i(\mathcal{L}).$$

The first inequality uses Hadamard's inequality and the second inequality invokes Minkowski's second theorem $\Pi_j \lambda_j(\mathcal{L}) \leq n^{n/2} \det(\mathcal{L})$. Then

$$\|w(M_h - I)\| = \left\| \sum_i \alpha_i (h(v_i) - v_i) \right\| \leq \sum_i |\alpha_i| \cdot \|h(v_i) - v_i\| \leq n \cdot \frac{n^n}{\sqrt{i} \lambda_i(\mathcal{L})} \cdot \beta \|v_i\| \leq n^{n+1} \beta.$$

This implies that $\|M_h - I\|_\infty \leq \beta^{(1)}$.

Next, by choosing $W := M_h$, we have $\|W - I\|_2 \leq \beta^{(2)} := \sqrt{n} \beta^{(1)} = n^{n+3/2} \beta$, and $M_{\mathcal{L}'} = M_{\mathcal{L}} W$ where $M_{\mathcal{L}'}$ (resp. $M_{\mathcal{L}}$) are matrices for the choice of basis of $\mathcal{L}'$ (resp. $\mathcal{L}$) that corresponds to M_h (i.e. $M_{\mathcal{L}'} = M_{\mathcal{L}} M_h$).

Then, since $\beta^{(2)} < (20\sqrt{n}|\Delta|)^{-1}$, W is necessarily diagonal (see Claim 1), and hence $M_{\mathcal{L}'} = M_{\mathcal{L}} e^{\text{diag}(a_i)}$ with $\|e^{\text{diag}(a_i)} - I\|_2 \leq \beta^{(2)}$. This implies that²

$$\|a\|_E = \|\text{diag}(a_i)\|_2 \leq 5\beta^{(2)} < \frac{1}{4\sqrt{n}|\Delta|}$$

when $\beta < \frac{1}{2n^{n+2}|\Delta|}$, and hence since $\|\text{diag}(a_i)\|_2 \geq \text{dist}(\mathcal{L}, \mathcal{L}')$, it contradicts the hypothesis that $\text{dist}(\mathcal{L}, \mathcal{L}') \geq r$. □

The following claim is taken from an unpublished version of [16].

Claim 1 (Sections E.2 and E.3 of [16]). *Let $\mathcal{L}$ (resp. $\mathcal{L}'$) be E -ideals of norm 1 admitting a basis represented by the matrix $M_{\mathcal{L}}$ (resp. $M_{\mathcal{L}'}$) satisfying $M_{\mathcal{L}'} = M_{\mathcal{L}} W$ for some matrix W . If $\|W - I\| < (2\sqrt{n}|\Delta|)^{-1}$, then $W = \text{diag}(z)$ for some $z \in E$.*

Proof. For completeness, we reproduce the proof of this statement as it is presented in [16]. The matrix W is of the form $\text{diag}(z)$ if and only if it commutes with all matrices of the form $\text{diag}(z)$. To check that, it suffices to show that $M_{\mathcal{L}}(W \text{diag}(\omega_j) - \text{diag}(\omega_j)W) = 0$ where $(\omega_k)_{k \leq n}$ is an integral basis of $\mathcal{O}$. Indeed both the ω_k and the rows $b_1, \dots, b_n$ of $M_{\mathcal{L}}$ are linearly independent. We can assume that $\|\omega_j\|_E \leq \lambda_n(\mathcal{O})$ and $\|b_j\|_E \leq \lambda_n(\mathcal{L})$. Moreover, we know that

$$\lambda_1(\mathcal{L}') \geq \sqrt{n} \mathcal{N}(\mathcal{L}')^{1/n}, \quad \lambda_n(\mathcal{O}) \leq \sqrt{n|\Delta|}, \quad \lambda_n(\mathcal{L}) \leq \sqrt{n|\Delta|} \mathcal{N}(\mathcal{L})^{1/n}$$

. Therefore, since $\mathcal{N}(\mathcal{L}), \mathcal{N}(\mathcal{L}') = 1$, we get

$$\forall k, \|b_k(W \text{diag}(\omega_j) - \text{diag}(\omega_j)W)\| \leq 2\|b_k\|_2 \|W - I\|_2 \|\omega_j\|_E < \sqrt{n} \mathcal{N}(\mathcal{L}')^{1/n} \leq \lambda_1(\mathcal{L}').$$

Since $M_{\mathcal{L}'} = M_{\mathcal{L}} W$, each $b_k(W \text{diag}(\omega_j) - \text{diag}(\omega_j)W)$ is a vector of $\mathcal{L}'$, therefore they have to be 0. □

Proof of Theorem 3. We need to show that there are $r, \varepsilon > 0$ such that

$$d_{G/L}(x, y) \geq r \Rightarrow |\langle f(x) | f(y) \rangle| < \varepsilon,$$

²Let $A = \text{diag } a_i$. Observe that $\|A\|_2 \leq 1$ and in this case $\sum_{k=2}^{\infty} \frac{\|A^k\|_2}{k!} \leq \|A\|_2 \cdot \sum_{k=2}^{\infty} \frac{1}{k!} = (e - 2)\|A\|_2$. Hence $\|e^A - I\|_2 \geq \|A\|_2 - \sum_{k=2}^{\infty} \frac{\|A^k\|_2}{k!} \geq 0.2\|A\|_2$.

where $d_{G/L}$ is the regular Euclidean distance in G/L , i.e. $d_{G/L}(x, y) = \min_{u \in L} \|x - y - u\|$. Let $\mathcal{L} = f_c(x)$ be the lattice corresponding to x and $\mathcal{L}' = f_c(y)$ be the one corresponding to y . With the notations of Definition 5, whenever $\Lambda \subsetneq \mathcal{L}$ (and $\Lambda' \subsetneq \mathcal{L}'$), we necessarily have $|\langle f(x)|f(y) \rangle| \leq 3/4$. Hence, by contraposition, we assume that $|\langle f(x)|f(y) \rangle| > 3/4$ (which implies $\Lambda = \mathcal{L}$ and $\Lambda' = \mathcal{L}'$), and we prove that this implies that $d_{G/L}(x, y)$ must be less than a certain bound r .

First, Lemma 3 implies that there is $\text{diag}(a_i)_{i \leq n}$ (in the sense of the diagonal matrices discussed in Lemma 1) such that $M_{\mathcal{L}'} = M_{\mathcal{L}}W$ for $W = e^{\text{diag}(a_i)}$ where $\|a\| \leq \frac{1}{4\sqrt{n}|\Delta|}$. This means that the matrix distance $\text{dist}(\mathcal{L}, \mathcal{L}')$ is necessarily less than $\frac{1}{4\sqrt{n}|\Delta|}$.

Next, we want to prove that if $\text{dist}(\mathcal{L}, \mathcal{L}') = \|A\|$ for some $A \in \text{GL}_n(\mathbb{R})$ with $M_{\mathcal{L}'} = M_{\mathcal{L}}e^A$, then A is necessarily of the form $\text{diag}(a'_i)$. We know that $\|A\| \leq \|\text{diag}(a_i)\| \leq \frac{1}{4\sqrt{n}|\Delta|}$. Moreover, for all A close to the zero matrix, the expansion of the matrix exponential tells us that

$$\|e^A - I\| = \left\| \sum_{k \geq 1} \frac{A^k}{k!} \right\| \leq \|A\| \sum_{k \geq 0} \frac{\|A\|^k}{k!} \leq (e - 1)\|A\| < 2\|A\|.$$

Hence $\|e^A - I\| < \frac{1}{2\sqrt{n}|\Delta|}$ and we can apply Claim 1 to argue that e^A is diagonal. Therefore, Since all e^A with $M_{\mathcal{L}'} = M_{\mathcal{L}}e^A$ and $\text{dist}(\mathcal{L}, \mathcal{L}') = \|A\|$ must be diagonal, we have that the matrix distance satisfies $\text{dist}(\mathcal{L}, \mathcal{L}') = \|a\|$ for some a with $M_{\mathcal{L}'} = M_{\mathcal{L}}e^{\text{diag}(a)}$ (where diag of matrices in $\mathbb{R}^{n \times n}$ is still understood as in Lemma 1).

In terms of E -lattices, this means that $\mathcal{L}' = \phi(a^G) \cdot \mathcal{L}$ where a^G is an element of G corresponding to a . To construct such an element, we first notice that $\det(e^{\text{diag}(a)}) = 1$, which means that the element $x^a \in \mathbb{R}^{n_1} \times \mathbb{C}^{n_2}$, corresponding to $e^{\text{diag}(a)}$ satisfies $|x_1^a| = \frac{1}{\prod_{i=2}^{n_1} |x_i^a| \prod_{i=n_1+1}^{n_2} |x_i^a|^2}$. We can therefore follow the construction of elements of G from S -units by treating $x^a \in \mathbb{R}^{n_1} \times \mathbb{C}^{n_2}$ as if it were in U_S with all valuations according to primes in S being 0. (i.e. all coordinates of a^G according to $\mathbb{Z}^s$ are set to 0). Since $M_{\phi(a^G)} = e^{\text{diag}(a)}$ is close to the identity matrix, we notice that this construction also directly implies that the real entries of $\phi(a^G) = x^a$ are close to 1, i.e. they are positive, and therefore all entries of a^G according to $\mathbb{Z}_2^{n_1}$ are zero. Moreover, each diagonal block Ξ_i corresponding to a complex coordinate of $\phi(a^G) = x^a$ is close to the identity block:

$$\Xi_i = \begin{pmatrix} \Re(x_i^a) & -\Im(x_i^a) \\ \Im(x_i^a) & \Re(x_i^a) \end{pmatrix} \sim \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}.$$

More specifically, since $\|e^{\text{diag}(a)} - I\| < \frac{1}{2\sqrt{n}|\Delta|}$, we know that

- $|\Im(x_i^a)| \leq \frac{1}{2\sqrt{n}|\Delta|}$
- $|\Re(x_i^a) - 1| \leq \frac{1}{2\sqrt{n}|\Delta|}$.

Hence, if $\theta_i \in \mathbb{R}/\mathbb{Z}$ satisfies $x^a = |x^a|e^{2i\pi\theta_i}$, we have

$$|\theta_i| \leq \frac{\pi}{2} |\sin(\theta_i)| = \frac{\pi}{2} \frac{|\Im(x_i^a)|}{|\Re(x_i^a)|} \leq \pi |\Im(x_i^a)| \leq \frac{\pi}{2\sqrt{n}|\Delta|}$$

Let $r = \frac{\pi}{2|\Delta|}$. We have $\|a^G\| \leq r$. Hence $d_{G/L}(x, y) \leq r$. This proves by contraposition that if $d_{G/L}(x, y) \geq r$, then $|\langle f(x)|f(y) \rangle| < \varepsilon = 3/4$. □

6 Lipschitz property of $(\mathbf{y}, \mathbf{v}) \mapsto |\phi(\mathbf{y})\mathcal{O} \prod_{\mathbf{p} \in S} \mathfrak{p}^{-v_i}\rangle$

Proposition 3 (Lipschitz property of f). *There is $a > 0$ such that*

$$||f(x)\rangle - |f(y)\rangle| < a \cdot d_{G/L}(x, y)$$

Proof. Let $z \in G$ such that $z = x - y - u$ where $u \in L$ is such that $d_{G/L}(x, y) = \|x - y - u\|$. If one of the components of z according to $\mathbb{Z}_2^{n_1}$, or $\mathbb{Z}^s$ is non-zero, then $d_{G/L}(x, y) \geq 1$, and then by the triangle inequality $|\langle f(x) | f(y) \rangle| \leq 2 \leq 2d_{G/L}(x, y)$. Now we assume that all components of z according to $\mathbb{Z}_2^{n_1}$ and $\mathbb{Z}^s$ are zero. In particular, this means that $\mathcal{L} = (e^z)\mathcal{L}'$ where $\mathcal{L} = f_c(x)$, $\mathcal{L}' = f_c(y)$, and $z \in E$ correspond to the canonical mapping of the components of z according to $\mathbb{R}^{n_1+n_2-1} \times (\mathbb{Z}_2)^{n_1} \times (\mathbb{R}/(\frac{1}{n^2}\mathbb{Z}))^{n_2}$. Therefore, we have $M_{\mathcal{L}} = M_{\mathcal{L}'} \cdot e^{\text{diag}(z)}$, and thus:

$$\begin{aligned} d_{G/L}(x, y) &= \|z\| \geq \|\phi(z)\| = \|\text{diag}(\phi(z))\|_2 \\ &\geq \inf\{\|A\|_2 : M_{\mathcal{L}} = M_{\mathcal{L}'} \cdot e^A\} \\ &\geq a_0 ||f(x)\rangle - |f(y)\rangle| \quad \text{by [16, Th. D.4]} \end{aligned}$$

Then we obtain the desired result with $a = \max\{2, 1/a_0\}$. $\square$

We have demonstrated the HSP property of f . We will now use this to derive the HSP property of $\tilde{f}$ which is a function from $\tilde{G}$ to $\mathcal{H}$ obtained from f .

7 An HSP oracle on $\mathbb{R}^m$

In the previous sections we described an oracle $f : G \rightarrow \mathcal{H}$ which satisfies the HSP properties of Definition 2 (in particular: pseudoinjectivity and Lipschitz property). We now show how to construct an oracle over $\mathbb{R}^m$ that hides the S -unit groups and that inherits the HSP properties of f . The control group G can be seen as the projection of $\tilde{G} = \mathbb{R}^k \times \mathbb{Z}^l$ where

- $k = n_1 + 2n_2 - 1$.
- $l = n_1 + s$.

We denote by $\gamma : \tilde{G} \rightarrow G$ the projection map, and by $\tilde{L} \subseteq \tilde{G}$ the pre-image of L by γ .

Definition 6 (Oracle on $\tilde{G}$). *We define $\tilde{f} : \tilde{G} \rightarrow \mathcal{H}$ by*

$$\tilde{f}(\tilde{x}) = f \circ \gamma(\tilde{x}).$$

We have the following diagram:

$$\begin{array}{ccccc} \mathbb{R}^m & \longleftarrow & \tilde{G} = \mathbb{R}^k \times \mathbb{Z}^l & \xrightarrow{\gamma} & G = \mathbb{R}^{n_1+n_2-1} \times \mathbb{Z}_2^{n_1} \times (\mathbb{R}/\mathbb{Z})^{n_2} \times \mathbb{Z}^s \\ & \swarrow \tilde{f} & \downarrow \tilde{\alpha} & & \downarrow \alpha \\ \mathcal{H} & & \tilde{G}/\tilde{L} & \xrightarrow{\pi} & G/L & \searrow f=f_q \circ f_c & \mathcal{H} \end{array}$$

We proceed by first showing that $\tilde{f}$ satisfies the HSP properties, and then we use techniques from [15, Th. 6.1] and [16, Sec. F] to derive an oracle on $\mathbb{R}^m$ that satisfies the HSP properties and that hides the S -unit group.

7.1 HSP properties of the oracle on $\tilde{G}$

In this section, we show that $\tilde{f}$ is an a, r, ε -oracle on $\tilde{G}$. Following the framework of [15, 16], we use the following distance on $\tilde{G}/\tilde{L}$.

Definition 7 (Distance on $\tilde{G}/\tilde{L}$). *Let $\tilde{x}, \tilde{y} \in \tilde{G}$. We define $d_{\tilde{G}}(x, y) = \|x - y\|$ if $x - y$ does not have any non-zero components on $\mathbb{Z}^l$, and $d_{\tilde{G}}(x, y) = \infty$ otherwise. Then*

$$d_{\tilde{G}/\tilde{L}} = \inf_{\tilde{u} \in \tilde{L}} d_{\tilde{G}}(x, y + \tilde{u}).$$

Proposition 4 (Lipschitz property of $\tilde{f}$). *Assume f is an a, r, ε -oracle. Then*

$$\forall \tilde{x}, \tilde{y} \in \tilde{G}, \quad \|\tilde{f}(\tilde{x})\rangle - \tilde{f}(\tilde{y})\rangle\| \leq a \cdot d_{\tilde{G}/\tilde{L}}(\tilde{x}, \tilde{y}).$$

Proof. Suppose $d_{\tilde{G}/\tilde{L}} = \infty$, then the inequality holds trivially. Otherise, Let $\tilde{u} \in \tilde{L}$ such that $d_{\tilde{G}/\tilde{L}}(\tilde{x}, \tilde{y}) = \|\tilde{x} - \tilde{y} - \tilde{u}\|$. In particular, all coordinates of $\tilde{z} := \tilde{x} - \tilde{y} - \tilde{u}$ with respect to $\mathbb{Z}^l$ are 0. Let $u = \gamma(\tilde{u})$, $x = \gamma(\tilde{x})$, and $y = \gamma(\tilde{y})$. We have

$$\|\tilde{x} - \tilde{y} - \tilde{u}\| \geq \|x - y - u\| \geq \min_{u \in L} \|x - y - u\| = d_{G/L}(x, y).$$

Hence $a \cdot d_{\tilde{G}/\tilde{L}}(\tilde{x}, \tilde{y}) \geq a \cdot d_{G/L}(x, y) \geq \|\tilde{f}(\tilde{x})\rangle - \tilde{f}(\tilde{y})\rangle\|$ □

Proposition 5 (Pseudoinjectivity of $\tilde{f}$). *Assume f is an a, r, ε -oracle for $r \ll 1$. Then*

$$d_{\tilde{G}/\tilde{L}}(\tilde{x}, \tilde{y}) \geq r \Rightarrow \left| \langle \tilde{f}(\tilde{x}) | \tilde{f}(\tilde{y}) \rangle \right| < \varepsilon$$

Proof. Let $x = \gamma(\tilde{x})$, $y = \gamma(\tilde{y})$, and $u \in L$ such that $d_{G/L}(x, y) = \|x - y - u\|$. If $z = x - y - u$ has no component on $\mathbb{Z}_2^{n_1}$ or $\mathbb{Z}^s$, then $d_{\tilde{G}/\tilde{L}}(\tilde{x}, \tilde{y}) = d_{G/L}(x, y)$ and therefore, if $d_{\tilde{G}/\tilde{L}}(\tilde{x}, \tilde{y}) \geq r$, then

$$|\langle \tilde{f}(\tilde{x}) | \tilde{f}(\tilde{y}) \rangle| = |\langle f(x) | f(y) \rangle| \leq \varepsilon.$$

On the other hand, if for such a u , we have components on $\mathbb{Z}_2^{n_1}$ or $\mathbb{Z}^s$, then either $d_{\tilde{G}/\tilde{L}}(\tilde{x}, \tilde{y}) = d_{\tilde{G}}(\tilde{x}, \tilde{y} + \tilde{u}) = \infty$, where $\tilde{u} \in \tilde{L}$ is the corresponding preimage, or $d_{\tilde{G}/\tilde{L}}(\tilde{x}, \tilde{y}) \geq 1$. So we only know that $d_{\tilde{G}/\tilde{L}}(\tilde{x}, \tilde{y}) \geq d_{G/L}(x, y)$ in this case. However, we also have that $d_{G/L}(x, y) \geq 1$ because of the integer components. Since $r \ll 1$ we necessarily have $d_{G/L}(x, y) \geq r$, and therefore $|\langle \tilde{f}(\tilde{x}) | \tilde{f}(\tilde{y}) \rangle| = |\langle f(x) | f(y) \rangle| \leq \varepsilon$. □

7.2 An HSP oracle on $\mathbb{R}^m$

Assume we have an $\tilde{a}, \tilde{r}, \tilde{\varepsilon}$ -oracle $\tilde{f}$ that hides U_S on $\tilde{G} = \mathbb{R}^k \times \mathbb{Z}^l$. Following [15, Th. 6.1] and [16, Sec. F] we derive an oracle $g : \mathbb{R}^m \rightarrow \mathcal{H}$ for $m = k + l$ defined by

$$|g(\mathbf{x}, y_1, \dots, y_l)\rangle = \sum_{z_1, \dots, z_l \in \{0, 1\}} \left(\bigotimes_{j=1}^l |\psi(y_j, z_j)\rangle \right) \otimes |\tilde{f}(\mathbf{x}, s(y_1, z_1), \dots, s(y_l, z_l))\rangle,$$

where $s(y, z) = \lfloor y/\lambda \rfloor + z$, and $|\psi(y, z)\rangle = \cos(\frac{\pi t}{2})|\text{str}_\nu(t)\rangle$ with $t = y/\lambda - s(y, z)$.

Theorem 4 (Theorem 6.1 of [15]). *If $\tilde{f}$ is an $\tilde{a}, \tilde{r}, \tilde{\varepsilon}$ -oracle, then g is an a', r', ε' -oracle with the following identities:*

$$\begin{aligned} a'^2 &= \tilde{a}^2 + l \left(\frac{\pi}{2\nu\lambda} (1 + \nu) \right)^2 \\ r'^2 &= \tilde{r}^2 + l(2\nu\lambda)^2 \\ \varepsilon' &= \tilde{\varepsilon}. \end{aligned}$$

7.3 Concrete parameters for the $\mathbb{R}$ -grid

Finally, we need to bound the first minima and the fundamental volume of the lattice of S -units. In the following, we show that these values have polynomial size with respect to the input. To bound the first minima of $U_S \subseteq G$ and the volume of G/U_S (which are preserved by the embedding of U_S into $\tilde{G}$), we rely on an analogue of Dirichlet unit theorem that applies to S -units. The classical results are known for the case where the lattice of S -units is embedded in $\mathbb{R}^{r+|S|}$ (where r is the rank of the unit group of $\mathcal{O}$) via the logarithm embedding

$$\text{Log}(\alpha) := \left(\log(|\alpha|_1), \dots, \log(|\alpha|_r), \log(|\alpha|_{\mathfrak{p}_1}), \dots, \log(|\alpha|_{\mathfrak{p}_{|S|}}) \right),$$

where $|\alpha|_j := |\sigma_j(\alpha)|$ and $|\alpha|_{\mathfrak{p}_j} := p_j^{-e_j v_{\mathfrak{p}_j}(\alpha)}$. In this case, we know from [21, Lem. 2] that $\|\text{Log}(\alpha)\|_\infty \geq \frac{\log(n)}{6n^4}$ where $\|v\|_\infty$ denote the usual infinity norm on the vector v , and

$$\text{Vol} \left(\mathbb{R}^{r+|S|} / \text{Log}(U_S) \right) \leq \left(300 \log(P) \sqrt{|\Delta|} \left(\frac{e}{2} \log(|\Delta|) \right)^{n-1} \right)^{|S|+r-\frac{n}{2}},$$

where $P = \max_j \mathcal{N}(\mathfrak{p}_j)$ (see [21, Sec. 2]).

Proposition 6. *The first minima of $U_S \subseteq G$ satisfies $\lambda_1(U_S) \geq \frac{\log(n)}{6n^4}$ where the norm on elements of G is defined by*

$$\|(z, v_1, \dots, v_{|S|})\| := \sqrt{\sum_j z_j^2 + \sum_j |v_j| e_j \log(p_j)}.$$

Moreover, the volume of the lattice of S -units satisfies

$$\text{Vol}(G/U_S) \leq \frac{2^{n_1}}{\log(2)^{|S|}} \left(300 \log(P) \sqrt{|\Delta|} \left(\frac{e}{2} \log(|\Delta|) \right)^{n-1} \right)^{|S|+r-\frac{n}{2}},$$

where $P = \max_j \mathcal{N}(\mathfrak{p}_j)$.

Proof. Let $((z_j), (v_k)) \in G$ corresponding to an S -unit α . We immediately see that

$$\|((z_j), (v_k))\| \geq \|\text{Log}(\alpha)\|_\infty,$$

which proves the lower bound on $\lambda_1(U_S)$.

To compute an upper bound on the volume of G/U_S , we follow the same approach as [16]. First, we consider the exact sequence $0 \rightarrow \mathbb{Z}^{n_1} \times (\mathbb{R}/\mathbb{Z})^{n_2} \rightarrow G \rightarrow \mathbb{R}^{m_1+n_2-1} \times \mathbb{Z}^s \rightarrow 0$. Let

$\mu(K)$ be the group of torsion units, and $L_S \subseteq \mathbb{R}^{n_1+n_2-1} \times \mathbb{Z}^s$ be the rank- n_1+n_2+s-1 -lattice that is the projection of U_S . Then we have the exact sequence

$$0 \rightarrow (\mathbb{Z}^{n_1} \times (\mathbb{R}/\mathbb{Z})^{n_2})/\mu(K) \rightarrow G/U_S \rightarrow (\mathbb{R}^{n_1+n_2-1} \times \mathbb{Z}^s)/L_S.$$

Hence $\text{Vol}(G/U_S) = \text{Vol}((\mathbb{Z}^{n_1} \times (\mathbb{R}/\mathbb{Z})^{n_2})/\mu(K)) \text{Vol}((\mathbb{R}^{n_1+n_2-1} \times \mathbb{Z}^s)/L_S)$

The volume of $(\mathbb{R}^{n_1+n_2-1} \times \mathbb{Z}^s)/L_S$ is equal to the absolute value of determinant of the matrix of a basis of L_S . Let $(\alpha_j)_{j \leq r+|S|}$ be a minimal generating set for $U_S/\mu(K)$. Its matrix M with respect to the embedding in $\mathbb{R}^r \times \mathbb{Z}^{|S|}$ is related to the matrix $M' := (\text{Log}(\alpha_j))$ by the relation $M = D \cdot M'$ where

$$D = \begin{pmatrix} 1 & (0) & & & \\ & \ddots & & & \\ & (0) & 1 & & \\ & & & 1/e_1 \log(p_1) & (0) \\ & & & \ddots & \\ & (0) & & & 1/e_{|S|} \log(p_{|S|}) \end{pmatrix}.$$

We therefore have

$$\begin{aligned} \text{Vol}((\mathbb{R}^{n_1+n_2-1} \times \mathbb{Z}^s)/L_S) &= \det(M) = \prod_j \frac{1}{e_j \log(p_j)} \det(M') \\ &= \prod_j \frac{1}{e_j \log(p_j)} \text{Vol}(R^{r+|S|}/\text{Log}(U_S)). \end{aligned}$$

Additionally, we have $\text{Vol}((\mathbb{Z}^{n_1} \times (\mathbb{R}/\mathbb{Z})^{n_2})/\mu(K)) = \frac{2^{n_1}}{|\mu(K)|}$, therefore, by using the upper bound on $\text{Vol}(R^{r+|S|}/\text{Log}(U_S))$, we get

$$\text{Vol}(G/U_S) \leq \frac{2^{n_1}}{\log(2)^{|S|}} \left(300 \log(P) \sqrt{|\Delta|} \left(\frac{e}{2} \log(|\Delta|) \right)^{n-1} \right)^{|S|+r-\frac{n}{2}}.$$

□

We are left to invoke the efficient HSP algorithm on $\mathbb{R}^m$. We recall its performance below.

Theorem 5 (Theorem 6.2 [16]). *Let f be an HSP instance with parameters $((a, r, \varepsilon), \lambda, d)$ such that $r \leq \frac{1}{36}\lambda$ and $\varepsilon \leq 1/2$. Let $0 < \eta \leq \lambda_1(L^*)$ and $0 < \mu < 1$ be a precision parameter and error parameter respectively. There is a quantum algorithm that makes $K = O(\log d + \log(1/\mu) + m \log a)$ oracle calls to f and generates K vectors $\tilde{u}^{(1)}, \dots, \tilde{u}^{(K)} \in \mathbb{R}^m$ having the following properties except with error probability at most μ :*

1. $\|\tilde{u}^{(i)}\| \leq \frac{3a}{\pi}$ for $i = 1, \dots, K$;
2. For each i , $\|\tilde{u}^{(i)} - u^{(i)}\| \leq \eta$ for some $u^{(i)} \in L^*$;
3. $u^{(1)}, \dots, u^{(K)}$ generate L^* .

The quantum circuit uses $O(Km(\log \frac{aK}{\eta\mu})^2)$ quantum gates on $O(m \log \frac{aK}{\eta\mu})$ qubits.

The above allows us to derive precise polynomial dependencies in the input parameters. For this, one needs to specify the cost of running the LLL algorithm in superposition. Generic transformations exist to turn a classical algorithm into a reversible quantum computation [3]. To facilitate a comparison with the recent prepring of de Boer and Felderhoff, we specialize our result to the case of the quantum LLL algorithm of [33, Eq 7,8].

Corollary 2. *Let K be a number field of degree n , and discriminant Δ . Let S be a set of prime ideals of K . The gate cost of each call to the oracle to compute the S -unit group is in*

$$\tilde{O}((n + |S|)^{5.5}(n + \log |\Delta|)^{5.5}n^{17.5}(n^2 + \log |\Delta|)^{3.5}).$$

Meanwhile, the qubit requirement of the oracle is in

$$\tilde{O}((n + |S|)^{2.5}(n + \log |\Delta|)^{2.5}n^{9.5}(n^2 + \log |\Delta|)^{3.5}).$$

Proof. To use Theorem 2, we need a bound on the bit size of the entries of the vectors given as input to the oracle. It is given by the qubit requirements of the quantum oracle, i.e. it is in

$$\tilde{O}\left(m(\log(a) + \log(K) + \log(\frac{1}{\eta}) + \log(\frac{1}{\mu}))\right).$$

With a precision of $2^{-O(d)}|\Delta|^{-O(1)}$, we obtain (using the notation of Theorem 2)

$$\log(v), \log(r) \in \tilde{O}((n + |S|)(n + \log |\Delta|)).$$

This means that the bound β on the bit size of the entries of the vectors given as input to the LLL algorithm satisfies

$$\beta \in \tilde{O}(n(n^2 + \log |\Delta|)(n + \log |\Delta|)(n + |S|)).$$

Using again $C_{\text{gates}}(k, \beta)$ and $C_{\text{qubits}}(k, \beta)$ to denote the gate cost (resp. the qubit cost) of running LLL in superposition, we have therefore a “classical oracle” gate cost in

$$\tilde{O}((n + |S|)^2(n + \log |\Delta|)^2C_{\text{gate}}(n^2, \beta))$$

Likewise, the qubit cost of the classical oracle is in

$$\tilde{O}((n + |S|)(n + \log |\Delta|)C_{\text{qubit}}(n^2, \beta))$$

If, similarly to [14], we use the methods of [33, Eq 7,8], then we can assume that $C_{\text{gates}}(k, \beta) = O(k^7\beta^{3.5})$ and $C_{\text{qubits}}(k, \beta) = O(k^4\beta^{1.5})$. The gate cost of our classical oracle is then in

$$\tilde{O}\left((n + |S|)^2(n + \log |\Delta|)^2n^{14}(n(n^2 + \log |\Delta|)(n + \log |\Delta|)(n + |S|))^{3.5}\right).$$

Meanwhile, the qubit cost of the oracle is then in

$$\tilde{O}\left((n + |S|)(n + \log |\Delta|)n^8(n(n^2 + \log |\Delta|)(n + \log |\Delta|)(n + |S|))^{1.5}\right).$$

The cost of the calculation of the quantum encoding of the output of the “classical oracle” is that of a matrix multiplication, which is negligible compared to the computation of the basis of the ideal itself. $\square$

8 Applications to other number theory problems

8.1 Recover an exact representation of the S -units

The solution of HSP is given to us as approximations of generators of the hidden subgroup. For many applications, an exact (and polynomially bounded) representation is preferable. Therefore, we process the solutions to the S -units problem classically to produce a compact representation of the generators of the S -unit group.

Definition 8 (Compact representation). *Let $l > 0$ be a constant, a compact representation of $\alpha \in \mathcal{O}$ with respect to the integral basis $(\omega_j)_{j \leq n}$ of $\mathcal{O}$ is a set of exact representations of polynomial size algebraic numbers γ_j satisfying $\alpha = \gamma_0 \gamma_1^l \cdots \gamma_k^{l^k}$, where k is polynomial in the size of the input.*

Biasse and Fieker [4, Sec. 5] described an efficient method based on [18, Alg. 7.53] to classically compute a compact representation of an algebraic number in polynomial time. These methods rely on the knowledge of an exact representation of the algebraic number we wish to represent (which is not the case here). A modification of [18, Alg. 7.53] using the approximation of the vector corresponding to an algebraic number yields a compact representation of that number.

Our algorithm for the compact representation of an S -unit takes as input $l > 0$ and a rational approximation (to an arbitrary polynomial precision q) of a vector of the form

$$(\log(|\alpha|_1, \dots, \log(|\alpha|_{n_1+n_2}), \theta, v_{\mathfrak{p}_1}(\alpha), \dots, v_{\mathfrak{p}_{|S|}}(\alpha)),$$

where α is an S -unit. We can assume that $\prod_j \mathfrak{p}_j^{v_{\mathfrak{p}_j}(\alpha)} \subseteq \mathcal{O}$. If not, we replace each $\log(|\alpha|_j)$ by

$$\left(\sum_{v_{\mathfrak{p}_k}(\alpha) < 0} |v_{\mathfrak{p}_k}(\alpha)| e_k \log(p_k) \right) + \log(|\alpha|_j)$$

(where $\mathcal{N}(\mathfrak{p}_k) = p_k^{e_k}$), thus calculating a compact representation of $\alpha \prod_{v_{\mathfrak{p}_k}(\alpha) < 0} p_k^{|v_{\mathfrak{p}_k}(\alpha)| e_k}$. From that, we can easily derive a compact representation of α . Then, we choose k_0 minimal such that $\frac{\log(|\alpha|_j)}{l^{k_0}} \leq \log(|\Delta|)$, initiate an ideal I to $\prod_j \mathfrak{p}_j^{\lfloor v_{\mathfrak{p}_j}(\alpha)/l^{k_0} \rfloor}$, and we compute rational approximations v_j of $(|\alpha|_j)^{1/l^k}$. Then at each step, I is replaced by I^l and we compute an LLL-reduced element δ_k of the ideal $C \subseteq \mathcal{O}$ such that $I^{-1} = \frac{1}{d_k} C$ for the scaled T_2 norm $T_{2,(v_j)_j}(\delta) := \sum |\delta|_i^2 \frac{v_j^2}{v^2}$ where $v := \sqrt[n]{\prod v_j}$. The ideal I is then replaced by $\beta_k I$ where $\beta_k := \frac{\delta_k}{d_k}$, and $v_j \leftarrow v_j \cdot |\beta_k|_j$. At every step k from k_0 to 0, we know that

- β_k has polynomial size,
- $\beta_{k_0}^{l^k} \cdots \beta_k(\alpha)^{\frac{1}{l^{k_0-k}}}$ has polynomial size,
- $I \subseteq \mathcal{O}$ and has polynomial size (i.e. $\log(I)$ is polynomial),
- $\prod_j v_j \geq \mathcal{N}(I) \geq 1$.

At the end of this process, we have polynomial size algebraic numbers $(\beta_j)_{j \leq k_0}$ such that $\beta_{-1} := \alpha \prod_k \beta_k^{l_k}$ has polynomial size. Finding β_{-1} is the main difference between our approach and that of [4, Sec. 5] and [18, Sec. 7] since we have no exact representation for α . We find the minimal $d > 0$ such that $\beta := d\beta_{-1} \in \mathcal{O}$ and from approximates of the $\log(|\beta_k|_j)$, $\log(|\alpha|_j)$, and the phase vector of each of the corresponding algebraic numbers, we find a rational approximation $\widehat{\beta} \in \mathbb{R}^m$ under the rule (4) with a polynomial number of bits of precision. Likewise, we can get approximations $\widehat{\omega}_j \in \mathbb{R}^m$ of the integral basis vectors ω_j , and solve the linear system (over the rationals) $\widehat{\beta} := \sum_j \frac{b_j}{c_j} \widehat{\omega}_j$. The nearest lattice point $\sum_j a_j \widehat{\omega}_j$ in $\sum_j \mathbb{Z} \widehat{\omega}_j$ can be retrieved if the precision is larger than n by using Babai's algorithm [1]. Then we know that $\beta = \sum_j a_j \omega_j$, and

$$\alpha = \frac{\beta_{-1}}{\beta_0} \left(\frac{1}{\beta_1} \right)^l \cdots \left(\frac{1}{\beta_{k_0}} \right)^{l_{k_0}}.$$

Algorithm 4 Compact representation

Input: Rational approximations of $\log(|\alpha|_j)$, phase vector of α and $v_j \geq 0$ such that $(\alpha) = \prod_j \mathfrak{p}_j^{v_j}$, $l > 0$, and approximations $\widehat{\omega}_j$ of an LLL-reduced integral basis of $\mathcal{O}$.

Output: Exact representation of $\gamma_0, \dots, \gamma_{k_0}$ such that $\alpha = \prod_k \gamma_k^{l_k}$.

- 1: $I \leftarrow \prod_j \mathfrak{p}_j^{\lfloor v_j(\alpha)/l^k \rfloor}$.
 - 2: Let k_j minimal such that $\frac{1}{l^k} \log |\beta_j|_i \leq \log \Delta$, $v_j \leftarrow \exp(l^{-k} \log |\alpha|_j)$
 - 3: **for** $0 \leq k \leq k_0$ **do**
 - 4: $B \leftarrow I^l$, $(w_j)_j \leftarrow (v_j^l)_j$.
 - 5: $w \leftarrow \sqrt[l]{\prod w_j}$ and $d_k \in \mathbb{Z}_{>0}$ such that $B^{-1} = \frac{1}{d_k} C$ for $C \subseteq \mathcal{O}$.
 - 6: Let δ be a 1st LLL-basis element of C with respect to $T_{2, (w_j/w)_j}(\delta) := \sum |\delta|_i^2 \frac{w_j^2}{w^2}$.
 - 7: $\beta_k \leftarrow \frac{\delta}{d_k}$, $I \leftarrow B\beta_k$, $(v_j)_{j \leq r+1} \leftarrow (w_j \cdot |\beta_k|_j)_{j \leq r+1}$.
 - 8: **end for**
 - 9: Let $\beta_{-1} = \alpha \cdot \prod_k \beta_k^{l_k}$
 - 10: Find an approximation $\widehat{\beta} \in \mathbb{R}^m$ of $d\beta_{-1}$ where $d \in \mathbb{Z}_{>0}$ is minimal such that $d\beta_{-1} \in \mathcal{O}$.
 - 11: Find $(a_j)_{j \leq n}$ such that $\sum_j a_j \widehat{\omega}_j$ is the closest vector to $\widehat{\beta}$ in $\sum_j \mathbb{Z} \widehat{\omega}_j$.
 - 12: $\beta_{-1} \leftarrow \frac{1}{d} \sum_j a_j \omega_j$.
 - 13: **return** $\frac{\beta_{-1}}{\beta_0}, \frac{1}{\beta_1}, \dots, \frac{1}{\beta_{k_0}}$.
-

Proposition 7. *Algorithm 4 is correct and returns a compact representation of the input α in polynomial time.*

Proof. The invariant properties on the size of the elements are deduced in the same way as in the proof of [4, Prop. 5.1]. The only important different is the way we compute an exact representation of β_{-1} . Babai's algorithm allows us to find in polynomial time a lattice element $\tilde{\beta}$ in $\widehat{\mathcal{L}} := \sum_j \mathbb{Z} \widehat{\omega}_j$ such that $d(\widehat{\beta}, \tilde{\beta}) \leq 2^n d(\widehat{\beta}, \widehat{\mathcal{L}})$. If the precision is larger than n , then the coefficients of $\tilde{\beta}$ on the basis $\widehat{\omega}_j$ are those of $\beta = d\beta_{-1}$ on the integral basis ω_j of $\mathcal{O}$. $\square$

8.2 Computation of class groups

Let $\mathcal{B} = \{\mathfrak{p}_1, \dots, \mathfrak{p}_N\}$ be a set of invertible prime ideals of an order $\mathcal{O}$ whose classes generate $\text{Cl}(\mathcal{O})$. We have a surjective morphism

$$\begin{array}{ccccc} \mathbb{Z}^N & \xrightarrow{\varphi} & \mathcal{I} & \xrightarrow{\pi} & \text{Cl}(\mathcal{O}) \\ (e_1, \dots, e_N) & \longrightarrow & \prod_i \mathfrak{p}_i^{e_i} & \longrightarrow & \prod_i [\mathfrak{p}_i]^{e_i}, \end{array}$$

and the class group $\text{Cl}(\mathcal{O})$ is isomorphic to $\mathbb{Z}^N / \ker(\pi \circ \varphi)$. Therefore, computing the class group boils down to computing $\ker(\pi \circ \varphi)$, which is the lattice of $(e_1, \dots, e_N) \in \mathbb{Z}^N$ such that $\mathfrak{p}_1^{e_1} \dots \mathfrak{p}_N^{e_N} = (\alpha)$ for some $\alpha \in K$. These α are S -units for $S = \mathcal{B}$, and the exponent vectors of a generating set of U_S give us a generating set for $\ker(\pi \circ \varphi)$ from which we derive $\text{Cl}(\mathcal{O})$.

The best unconditional bounds on $|S|$ are exponential in $\log(|\Delta|)$. As the complexity of the computation of the $|S|$ -unit group is polynomial in $|S|$, we cannot achieve a polynomial complexity unconditionally that way. However, under the Generalized Riemann Hypothesis (GRH), the classes of all prime ideals of $\mathcal{O}$ of norm up to $48 \log(|\Delta|)^2$ generate $\text{Cl}(\mathcal{O})$. The size of $S := \{\mathfrak{p} \subseteq \mathcal{O} \text{ prime} \mid \mathcal{N}(\mathfrak{p}) \leq 48 \log(|\Delta|)^2\}$ is polynomial in $\log(|\Delta|)$, and calculating the corresponding S -unit group is therefore polynomial in n and $\log(|\Delta|)$.

Algorithm 5 Ideal class group of $\mathcal{O}$

Input: $\mathcal{O}$

Output: $d_1, \dots, d_n$ such that $\mathcal{O} \simeq \mathbb{Z}/d_1\mathbb{Z} \oplus \dots \oplus \mathbb{Z}/d_n\mathbb{Z}$.

- 1: $S \leftarrow \{\mathfrak{p} \subseteq \text{prime} \mid \mathcal{N}(\mathfrak{p}) \leq 48 \log(|\Delta|)^2\}$.
 - 2: Compute the S -unit group U_S .
 - 3: Let $(\alpha_j, v_{j,1}, \dots, v_{j,|S|})_{j \leq r+|S|}$ be the generating set for U_S computed.
 - 4: $\text{diag}(d_1, \dots, d_n) \leftarrow \text{Smith Normal Form of } M = (v_{j,k})$.
 - 5: **return** $d_1, \dots, d_n$.
-

Proposition 8. *Under the Generalized Riemann Hypothesis, Algorithm 5 is correct and runs in polynomial time.*

Our work also has direct applications in computational number theory. Indeed, the S -unit group is a central object that can be used in a lot of algorithms. It usually is computed together with the so-called S -class group, which is the quotient of the group of ideals in the ring of S -integers by the subgroup of principal ideals. The S -class group can easily be derived from the ideal class group and an oracle for the PIP by quotienting the class group by extra relations. A description of this method can be found in Simon's PhD thesis [31, Chap. 1].

Another direct consequence of our work is that it directly implies a polynomial time algorithm for computing the relative class group and the relative unit group of an arbitrary extension of number fields. Algorithms for these tasks are already known [?][Ch. 7], but their run time is exponential in the degree of the fields. As for the S -class group, they also consist of using a complete set of relations for the ideal class group and of enriching it with new relations that are obtained by solving instances of the PIP.

8.3 Resolution of the principal ideal problem

Let $\mathfrak{a} \subseteq \mathcal{O}$ be an ideal of $\mathcal{O}$. We want our algorithm to run in polynomial time in the size of the input, that is $\log |\Delta|$, n , and $\log(\mathcal{N}(\mathfrak{a}))$ (which quantifies the size of $\mathfrak{a}$). The ideal $\mathfrak{a}$ is principal if and only if $\mathfrak{a} = (\alpha)$ for α an S -unit where S is the set of prime divisors of $\mathfrak{a}$. We calculate a generating set for the S -units, which gives us a generating set for all the possible principal ideals only divisible by elements of S . The resolution of a linear system tells us if $\mathfrak{a}$ belongs to this set, and if so, what is its generator.

Algorithm 6 Principal ideal problem

Input: $\mathcal{O}$ and an ideal $\mathfrak{a} \subseteq \mathcal{O}$.

Output: Decide if $\mathfrak{a}$ is principal and if so a compact representation of a generator α .

- 1: Factor $\mathfrak{a}$, let $S = \{\mathfrak{p}_1, \dots, \mathfrak{p}_k\}$ be the divisors of $\mathfrak{a}$.
 - 2: Compute the S -unit group $U_S = \mu \times \langle \varepsilon_1 \rangle \times \dots \times \langle \varepsilon_{r+|S|} \rangle$.
 - 3: Let $M = (m_{i,j})$ such that $\varepsilon_i = \prod_j \mathfrak{p}_j^{m_{i,j}}$.
 - 4: Solve $XM = \mathfrak{a}$ where $\mathfrak{a} = \prod_i \mathfrak{p}_i^{a_i}$.
 - 5: **return** compact representation of $\prod_i \varepsilon_i^{x_i}$ or “not principal” if the system has no solution.
-

8.4 Ideal class decomposition in $\text{Cl}(\mathcal{O})$

Under the GRH, the set of prime ideals

$$S := \{\mathfrak{p} \subseteq \mathcal{O} \text{ prime} \mid \mathcal{N}(\mathfrak{p}) \leq 48 \log(|\Delta|)^2\} \cup \{\mathfrak{p} \subseteq \mathcal{O} \text{ prime} \mid \mathfrak{p} \mid \mathfrak{a}\}$$

generate the ideal class group. Ideal class decomposition consists of finding exponents $x_1, \dots, x_s$ and $\alpha \in K$ such that

$$\mathfrak{a} = (\alpha) \mathfrak{p}_1^{x_1} \dots \mathfrak{p}_s^{x_s}.$$

We want our algorithm to run in polynomial time in the size of the input, that is $\log |\Delta|$, n , and $\log(\mathcal{N}(\mathfrak{a}))$ (which quantifies the size of $\mathfrak{a}$). Our strategy is the following:

1. Decompose $\mathfrak{a}$ as a product of prime ideals $\mathfrak{a} = \prod \mathfrak{q}$.
2. For each $\mathfrak{q}_j \notin \mathcal{B}, j \leq k$ in the decomposition of $\mathfrak{a}$, find $\beta_k \in K$ such that $\mathfrak{q} = (\beta_k) \cdot \prod_{\mathfrak{p}_j \in S} \mathfrak{p}_j^{x_{j,k}}$.
3. Deduce $\mathbf{v} \in \mathbb{Z}^N$ such that $\mathfrak{a} = \prod_k (\beta_k) \cdot \prod_{\mathfrak{p}_j \in \mathcal{B}} \mathfrak{p}_j^{v_j}$.

deciding if an input ideal $\mathfrak{a} \subseteq \mathcal{O}$ is principal, and if so, compute an element $\alpha \in \mathcal{O}$ such that $\mathfrak{a} = (\alpha)$. The first step consist of finding the prime ideal decomposition of $\mathfrak{a}$. Then we define S by

$$S := \{\mathfrak{p} \subseteq \mathcal{O} \text{ prime} \mid \mathcal{N}(\mathfrak{p}) \leq 48 \log(|\Delta|)^2\} \cup \{\mathfrak{p} \subseteq \mathcal{O} \text{ prime} \mid \mathfrak{p} \mid \mathfrak{a}\},$$

and compute the S -unit group. Then we deduce the solution to the principal ideal problem by performing a linear algebra step on the matrix of the valuations, as described in Section ??.

Proposition 9. *Under the Generalized Riemann Hypothesis, Algorithm 7 is correct and runs in polynomial time.*

Algorithm 7 Ideal class decomposition

Input: $\mathcal{O}$ and an ideal $\mathfrak{a} \subseteq \mathcal{O}$.

Output: Decide if $\mathfrak{a}$ is principal and if so a compact representation of a generator α .

- 1: Factor $\mathfrak{a}$.
 - 2: $S \leftarrow \{\mathfrak{p} \subseteq \text{prime} \mid \mathcal{N}(\mathfrak{p}) \leq 48 \log(|\Delta|)^2\}$.
 - 3: $S \leftarrow S \cup \{\mathfrak{p} \subseteq \mathcal{O} \text{ prime} \mid \mathfrak{p} \mid \mathfrak{a}\}$.
 - 4: $\mathbf{v} \leftarrow$ vector of valuations of $\mathfrak{a}$ according to S .
 - 5: Compute the S -unit group U_S .
 - 6: Let $(\alpha_j, v_{j,1}, \dots, v_{j,|S|})_{j \leq r+|S|}$ be the generating set for U_S computed.
 - 7: Compute a compact representation of the α_j .
 - 8: Find $U \in GL_{r+|S|}(\mathbb{Z})$ and H such that $U \begin{pmatrix} H|0 \\ B|I \end{pmatrix}$ is the HNF of $(v_{j,k})$ and $I = I_m$.
 - 9: $\beta_j \rightarrow \prod_k \alpha_k^{U_{j,k}}$ in compact representation for $j \leq r + |S|$.
 - 10: $\mathbf{v} \rightarrow \mathbf{v}_1 + B\mathbf{v}_2$ where $\mathbf{v} = (\mathbf{v}_1, \mathbf{v}_2)$.
 - 11: **return** $\prod_k (\beta_k), \mathbf{v}$
-

8.5 Ray class groups

Our algorithms also directly imply a quantum algorithm for computing the ray class group of an arbitrary number field. The computation of the ray class group is an essential task in computational class field theory, and both classical and quantum algorithms have been described to solve this task. A classical method due to Cohen can be found in [?][3.2] and has an exponential run time with respect to the degree (but runs in subexponential time for classes of fixed degree number fields). A quantum algorithm was described by Eisenträger and Hallgren [?] with a polynomial run time in classes of fixed degree number fields. As for the aforementioned tasks, computing the ray class group essentially relies on subroutines for computing the ideal class group and solving the PIP, for which we provide polynomial time algorithms in arbitrary number fields. It also relies on algorithms for factoring ideals (which can be easily derived from Shor's factoring algorithm), and efficient methods for solving the discrete logarithm problem (which is also a well known consequence of Shor's work [30]).

8.6 Norm equations

Finally, our work allows us to describe polynomial time algorithms for solving relative norm equations of the form $\mathcal{N}_{L/K}(x) = \theta$ where L/K is an arbitrary Galois extension. Norm equations are an important example of Diophantine equations which are a major topic in number theory. The resolution of the Pell equation (for which there is a quantum algorithm [23]) can be seen as a special case where $L = \mathbb{Q}(\sqrt{\Delta})$, $K = \mathbb{Q}$ and $\theta = 1$ (when we restrict our attention to integer solutions). Solving norm equations in general is an important task in computational number theory. A classical method was described by Simon [31] (based on the work of Fieker [17] for Galois extensions) that solves general extensions in exponential time in the degree of the fields. For the Galois case, it simply uses the knowledge of the S -unit group and the relative class group, which we can provide in polynomial time for number fields of arbitrary degree. However, the general method

uses the Galois closure, whose degree can be exponential in the degree of the field, thus restricting the direct application of our work to arbitrary Galois extensions.

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